Adversarial Modality Interactions: Provenance-Aware Partial Information Decomposition for Strategically-Generated Multimodal Content in Real Estate

Joseph Firmansyah **Stephanie Chen** **Sophie Lin**
josephfn@mit.edu steph@mit.edu sl130@wellesley.edu

Abstract

Real-estate listings, like product reviews and many other forms of multimodal content, mix modalities curated by an agent with a stake in the outcome with modalities collected independently. Standard partial information decomposition (PID) of Williams and Beer [22] and Liang et al. [13] decomposes the joint information about a target into redundancy, uniqueness, and synergy, but treats all unique information as legitimate by construction. We introduce **adversarial modality interactions**, an extension of the PID lattice in which the unique information of a strategically-generated modality $X_{\mathcal{S}}$ further decomposes into a truthful part $U_{\mathcal{S}}^{\text{rel}}$ and an adversarial part $U_{\mathcal{S}}^{\text{adv}}$. We provide a three-source PID estimator under structural assumptions tied to which modalities the agent controls, validate it on a Gaussian data-generating process where the ground-truth atom is known in closed form (mean relative error of 0.43% across four buyer-rationality regimes), and apply it to 20,254 Boston-area listings with six modalities: listing text, listing photographs, Google Street View, satellite tiles (CLIP-encoded), plus tabular property attributes and Census-tract demographics (covariates). The standard 2-source unique atom on the price-gap target is 0.0027 nats; our 3-source $U_{\mathcal{S}}^{\text{adv}}$ is 0.0390 nats in-sample (out-of-sample 0.0121), $14.4\times$ larger, $z = 25.4$ above a 1000-permutation null, with a 95% cluster-bootstrap confidence interval of $[0.0345, 0.0426]$. Our one-dimensional supervised direction exceeds the information-theoretic upper bound on any 64-dimensional principal-component reduction of $X_{\mathcal{S}}$ by $\mathbf{3.4}\times$, which establishes that the adversarial atom cannot be recovered by any unsupervised method that ranks directions by variance. Within-building fixed effects yield $b = -1.24$ ($p = 0.008$), a broker-LOO instrument yields $b = -9.67$ ($F = 1497$), and AIPW, caliper matching, and a causal forest all agree across eleven specifications. A cross-modal text-photo synergy atom of 0.0123 nats ($z = 12.9$ over null) maps the result onto Liang's synergy concept, and a broker-LOO instrument for spatial peer effects identifies multi-sender Bayesian persuasion [4] at $\beta_{\text{peer}} = 1.77$ ($t = 8.4$), $4.5\times$ the OLS estimate.

1 Introduction

Consider a listing agent who describes a ground-floor Boston apartment as "a sun-drenched retreat with vibrant neighborhood energy." A buyer who checks Google Street View instead sees a north-facing unit beside a parking structure. Neither source alone produces the insight. The text offers no basis for skepticism; the image provides no claim to evaluate. The divergence between them is what carries information about whether the agent is overstating.

Real-estate listings are an ideal case for studying multimodal content with mixed provenance because they split cleanly along incentive lines. The agent curates the listing text and selects which photographs appear. The agent does not control Google Street View, satellite imagery, tax records, or Census-tract demographics. A standard analysis treats all six modalities as honest observations of the same property, fuses them, and predicts price. The dominant interaction taxonomies in multimodal learning support this view: partial information decomposition (PID) factors the joint information $I(X_{\mathcal{S}}, X_{\mathcal{O}}; Y)$ about a target Y into redundancy, uniqueness, and synergy [22, 21, 13], with all unique information assumed legitimate by construction. The taxonomy has no category for the gap between what an agent says and what an independent source shows.

A first observation suggests this matters in practice. We compare three fusion architectures on two targets, evaluated by 5-fold cross-validated R^2 on our 20,254-listing sample. On the log-price target, tabular features alone (beds, baths, square footage) reach $R^2 = 0.886$. Late fusion (five separate Ridge predictions averaged) drops to 0.503. Early fusion (concatenated CLIP features) recovers to 0.876. Visual modalities saturate against the tabular baseline; the multimodal contribution on log price is essentially zero. On the price gap, defined as $(p_{\text{list}} - p_{\text{sold}})/p_{\text{list}}$, the pattern inverts. Tabular alone reaches $R^2 = 0.075$. Late fusion drops to 0.058. Early fusion reaches 0.114. The visual contribution over tabular is eight times larger on the gap than on log price, and early fusion outperforms late fusion by 97% relative on the gap. The price gap, not log price, is the target where multimodal interaction structure becomes essential, and Section 7 returns to this with the full comparison.

This observation motivates the methodological question of the paper. In the standard 2-source PID decomposition, what is the structure of the unique-to-strategic atom $U_{\mathcal{S}}$ on the price gap? We propose that when one modality is strategically generated, its unique information further decomposes:

$$U_{\mathcal{S}} \;=\; U_{\mathcal{S}}^{\text{rel}} \;+\; U_{\mathcal{S}}^{\text{adv}}. \tag{1}$$

The truthful subcomponent $U_{\mathcal{S}}^{\text{rel}}$ captures additional honest content the agent observes (a renovation visible in a photo but not in Street View). The adversarial subcomponent $U_{\mathcal{S}}^{\text{adv}}$ captures claims that objective modalities contradict. Identification proceeds through structural assumptions about which modalities the agent controls, which sidesteps the underdetermination of continuous PID in the asymmetric case.

Preprint.

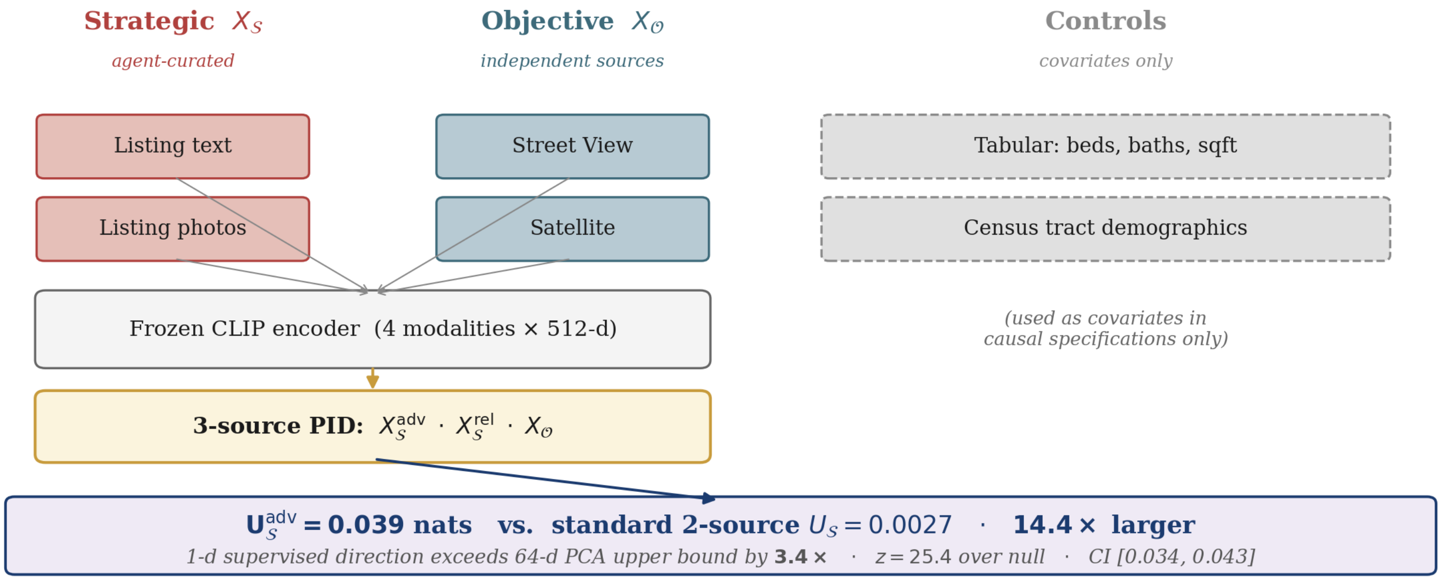

Figure 1: Pipeline overview. Six modalities, organized by provenance. Strategic modalities (agent-curated, warm) and objective modalities (independent, cool) are CLIP-encoded and feed the three-source PID decomposition. Tabular property attributes and Census-tract demographics serve as control covariates in causal specifications and do not feed the PID. The headline atom on the price-gap target is 0.039 nats, 14.4× the standard 2-source atom of 0.0027 nats. The 1-d supervised direction exceeds the 64-d PCA upper bound by 3.4×.

We make three contributions. **(1) Provenance-aware PID with synthetic calibration.** Section 4 gives the formal three-source decomposition with structural identification. Section 5 provides a supervised estimator and validates it on a Gaussian data-generating process where the ground-truth atom is known analytically; the estimator recovers ground truth at 0.43% relative error across four regimes. **(2) Empirical case with multi-angle causal identification.** On 20,254 Boston-area listings, the 3-source U_S^{adv} on the price gap is 0.0390 nats, 14.4× the standard 2-source atom of 0.0027. The same one-dimensional supervised direction exceeds the information-theoretic upper bound on any 64-dimensional PCA reduction of X_S by 3.4×, which is the strongest available evidence that supervision is information-theoretically necessary. Seven causal identification strategies across eleven specifications all point in the same direction. **(3) Multimodal-interaction extensions.** A cross-modal text-photo synergy atom of 0.0123 nats ($z = 12.9$) maps onto Liang's synergy concept and confirms that adversarial intent is coordinated across modalities. A broker-LOO instrument for spatial peer effects identifies multi-sender Bayesian persuasion [4] at $\beta_{\text{peer}} = 1.77$ ($t = 8.4$), 4.5× the OLS estimate.

2 Related Work

Multimodal interaction decomposition. Williams and Beer [22] introduce the PID redundancy lattice. Liang et al. [13] scale it to deep learning settings; Liang et al. [12] factorize representations into shared and unique parts via contrastive learning. Yu et al. [24] and Xin et al. [23] build mixture-of-experts architectures that exploit interaction structure. Zhao et al. [27] use normalizing flows for continuous PID estimation; Ma et al. [16] enforce Gaussianity with a closed-form conditional-independence regularizer. Both flag continuous PID as underdetermined and impose smoothness rather than provenance. Our extension along the provenance axis is orthogonal: we keep the lattice but split U_S into adversarial and truthful subatoms by structural assumption.

Information economics and multi-sender persuasion. Akerlof [1] formalizes how information asymmetry degrades market quality. Kamenica and Gentzkow [6] introduce Bayesian persuasion. Hossain et al. [4] extend the framework to multi-sender computational settings; the broker-pool structure in our data is a natural empirical instance. Levitt and Syverson [9] document broker-mediated information asymmetry in real-estate transactions specifically.

Multimodal real estate. Law et al. [8] use Street View and satellite imagery to improve hedonic models. Hasan et al. [3] learn joint embeddings across property images, text, and tabular features. Huang et al. [5] survey the area. All three treat modalities as neutral information sources collected from a single perspective; none partitions modalities by who controls them.

Multimodal foundation models. CLIP [18], ALBEF [10], and SigLIP [26] provide aligned vision-language embeddings used here as frozen base features. Tsai et al. [20] introduces cross-modal attention; Zadeh et al. [25] builds outer-product fusion; both serve as baselines in Appendix N. Liang et al. [15] argue multimodal contributions should be evaluated against properly-strong unimodal baselines; Liang et al. [14] derive bounds for multimodal interactions using disagreement between separately trained unimodal classifiers.

To our knowledge, this paper is the first to extend PID along the provenance axis. The closest prior work [13] decomposes joint information into redundancy, uniqueness, and synergy assuming all sources are neutral. We add an internal decomposition of U_S when one source is strategically generated, identified through structural assumptions about modality provenance rather than information-theoretic constraints alone.

3 Problem Statement

Each listing i has six modalities. Four are CLIP-encoded [18]: listing text ($x_i^{\text{text}} \in \mathbb{R}^{512}$), listing photographs mean-pooled across 15 to 40 shots ($x_i^{\text{photo}} \in \mathbb{R}^{512}$), four cardinal-heading Google Street View images mean-pooled ($x_i^{\text{sv}} \in \mathbb{R}^{512}$), and one Google Maps satellite tile

($x_i^{\text{sat}} \in \mathbb{R}^{512}$). The strategic block is $X_{\mathcal{S},i} = (x_i^{\text{text}}, x_i^{\text{photo}}) \in \mathbb{R}^{1024}$; the objective block is $X_{\mathcal{O},i} = (x_i^{\text{sv}}, x_i^{\text{sat}}) \in \mathbb{R}^{1024}$. Two additional objective sources serve as control covariates in causal specifications: tabular property attributes (beds, baths, square footage, list price) and Census-tract demographics (median income, percent college-educated, median home value). Outcomes are the log sale price $Y_i^{\text{price}} = \log p_i^{\text{sold}}$ and the price gap $Y_i^{\text{gap}} = (p_i^{\text{list}} - p_i^{\text{sold}})/p_i^{\text{list}}$, where negative values indicate sale above the asking price (a bidding war).

The standard 2-source PID of $I(X_{\mathcal{S}}, X_{\mathcal{O}}; Y)$ [22] factors the joint information into four nonnegative atoms:

$$I(X_{\mathcal{S}}, X_{\mathcal{O}}; Y) \;=\; R(X_{\mathcal{S}}, X_{\mathcal{O}}; Y) \;+\; U_{\mathcal{S}}(Y) \;+\; U_{\mathcal{O}}(Y) \;+\; S(X_{\mathcal{S}}, X_{\mathcal{O}}; Y), \tag{2}$$

where R is redundancy, $U_{\mathcal{S}}$ and $U_{\mathcal{O}}$ are the unique-to-strategic and unique-to-objective atoms, and S is synergy. Our objective is to further decompose $U_{\mathcal{S}} = U_{\mathcal{S}}^{\text{rel}} + U_{\mathcal{S}}^{\text{adv}}$ via a three-source PID with sources $(X_{\mathcal{S}}^{\text{adv}}, X_{\mathcal{S}}^{\text{rel}}, X_{\mathcal{O}})$, and to identify, estimate, and validate $U_{\mathcal{S}}^{\text{adv}}$ on the price-gap target.

4 Adversarial Modality Interactions

4.1 The provenance asymmetry

The asymmetry that drives identification is operational. The listing agent chooses listing text and selects which photographs to publish; the seller does not choose the angle of the Street View car or the resolution of the satellite tile. Under this asymmetry, $X_{\mathcal{S}}$ carries (i) honest information shared with $X_{\mathcal{O}}$ (redundancy), (ii) honest information unique to $X_{\mathcal{S}}$ (a renovation visible in a photo but not in Street View), and (iii) adversarial information unique to $X_{\mathcal{S}}$ (a north-facing unit beside a parking structure presented as a sun-drenched retreat). The standard 2-source PID atom $U_{\mathcal{S}}$ conflates (ii) and (iii).

4.2 Three-source decomposition and structural identification

Let $w_{\text{adv}} \in \mathbb{R}^{1024}$ with $\|w_{\text{adv}}\| = 1$ be a unit direction in strategic-modality space. Define the scalar adversarial projection and the residual:

$$X_{\mathcal{S}}^{\text{adv}} \;=\; \langle X_{\mathcal{S}},\, w_{\text{adv}} \rangle, \qquad X_{\mathcal{S}}^{\text{rel}} \;=\; X_{\mathcal{S}} \;-\; \langle X_{\mathcal{S}},\, w_{\text{adv}} \rangle \, w_{\text{adv}}. \tag{3}$$

With sources $(X_{\mathcal{S}}^{\text{adv}}, X_{\mathcal{S}}^{\text{rel}}, X_{\mathcal{O}})$ the three-source PID lattice has 18 atoms [22]. The atom of interest is the unique-to-adversarial atom, which we operationalize as the conditional mutual information

$$\text{cmi}_{\text{adv}} \;:=\; I\big(X_{\mathcal{S}}^{\text{adv}};\, Y^{\text{gap}} \,\big|\, X_{\mathcal{O}},\, X_{\mathcal{S}}^{\text{rel}}\big)\,. \tag{4}$$

When sources are jointly Gaussian and the outcome is linear-Gaussian, cmi_{adv} equals the Williams-Beer $I_{\min}$ unique atom $U_{\mathcal{S}}^{\text{adv}}$ on $X_{\mathcal{S}}^{\text{adv}}$ (Appendix D). For general distributions, cmi_{adv} is a lower bound on $U_{\mathcal{S}}^{\text{adv}}$. We use Gaussian estimates throughout the main paper and validate them against the non-parametric Kraskov-Stögbauer-Grassberger (k-nearest-neighbor) MI estimator [7] in Appendix F.

Structural identification. Let V denote latent value (the property's true worth) and A denote latent adversarial intent (the agent's strategic effort to overstate). We impose two structural conditions on w_{adv}:

(C1) Value orthogonality: $X_{\mathcal{S}}^{\text{adv}} \perp V$.

(C2) Objective orthogonality: $X_{\mathcal{S}}^{\text{adv}} \perp X_{\mathcal{O}}$.

Under (C1) and (C2), $X_{\mathcal{S}}^{\text{adv}}$ carries only information about strategic content uncorrelated with the property's value and uncorrelated with independent observations. Combined with the third constraint that $X_{\mathcal{S}}^{\text{adv}}$ predicts Y^{gap}, the direction w_{adv} is identified up to sign. In our data, (C1) and (C2) hold partially: $r = 0.23$ between $\hat{X}_{\mathcal{S}}^{\text{adv}}$ and the value-supervised projection of $X_{\mathcal{O}}$. We operate in a partial-identification regime and address this in Section 9.

4.3 The structural data-generating graph

Figure 2 encodes the conditions above as a directed acyclic graph. Three latents: effort E, value V, and adversarial intent A. Objective modalities load on value alone (the Street View car records the building as it is). Strategic modalities load on value, effort, and adversarial intent. Sale price reflects value after the buyer's discounting; the price gap reflects adversarial intent, the residual that survives the buyer's discounting. The graph operationalizes (C1) and (C2) as direct structural assumptions about which factors load on which modalities; it is the source of identification.

4.4 Linking the framework to the experiments

Each piece of the framework above corresponds to a specific empirical test in this paper, summarized in Table 1. The structural identification conditions (C1) and (C2) are tested directly in Appendix G; the formal estimator in Equation 4 is calibrated against analytic ground truth in Section 5.2 and Appendix E; the headline atom magnitude is reported in Section 8.1; the structural causal DAG is tested through within-property and within-building fixed effects in Section 8.3, which restricts variation to settings where V is approximately held fixed; partial identification is bounded by the doubly-robust AIPW analysis. The point of this subsection is to make the math-data correspondence explicit before reporting results.

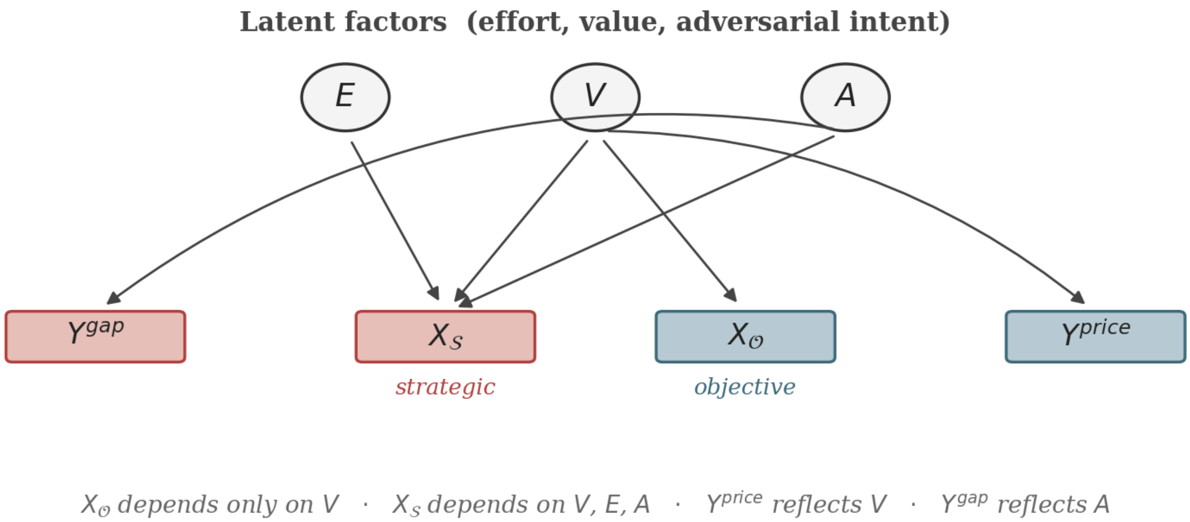

Figure 2: The strategic-objective causal graph. Objective modalities $X_{\mathcal{O}}$ load on value V alone; strategic modalities $X_{\mathcal{S}}$ load on value, effort E, and adversarial intent A. Sale price reflects value; price gap reflects adversarial intent. The adversarial latent A is identified by independence from V, $X_{\mathcal{O}}$, and Y^{price} while predicting Y^{gap}.

Table 1: Each framework component maps to a specific empirical test.

Framework piece	Empirical test	Result location
3-source PID decomposition (Eq. 3)	Headline atom magnitude	Section 8.1, Table 2
Identification condition (C1) $X_{\mathcal{S}}^{\text{adv}} \perp V$	Specificity ratio on Y^{price}	Section 8.1, Table 2
Identification condition (C2) $X_{\mathcal{S}}^{\text{adv}} \perp X_{\mathcal{O}}$	Cosine with value-supervised $X_{\mathcal{O}}$	Appendix G
Gaussian-MI equality (Eq. 4)	KSG validation vs. Gaussian MI	Appendix F
Closed-form atom (Eq. 7)	Synthetic recovery across γ regimes	Section 5.2, Fig. 11
Structural DAG (Fig. 2)	Within-property and within-building FE	Section 8.3, Fig. 6
A is a real causal driver	Broker-LOO IV; AIPW; causal forest	Section 8.3, Fig. 6
Multi-source PID lattice	Cross-modal synergy with null test	Section 8.4, Fig. 7
Multi-sender persuasion structure	Spatial peer-effect IV identification	Section 8.5, Fig. 8

5 Estimation

5.1 Supervised estimator

We fit w_{adv} by ridge regression of Y^{gap} on $X_{\mathcal{S}}$ with regularization $\alpha = 1.0$:

$$\hat{w}_{\text{adv}} \;=\; \arg\min_{w} \;\sum_{i} \big(Y_i^{\text{gap}} - w^\top X_{\mathcal{S},i}\big)^2 \;+\; \alpha \|w\|^2, \tag{5}$$

normalized to unit length. Five-fold out-of-fold projections yield the per-listing scalar $\phi_{\mathcal{S},i} = \langle X_{\mathcal{S},i}, \hat{w}_{\text{adv}} \rangle$, used throughout the empirical analysis. The five fold-trained directions have pairwise cosine averaging 0.82 in differentiated cities (Boston, Cambridge, Newton) and 0.74 in commodity cities (Arlington, Brookline, Braintree); see Appendix G. The conditional mutual information in Equation 4 is computed by Gaussian-MI closed-form expressions on $(X_{\mathcal{S}}^{\text{adv}}, X_{\mathcal{S}}^{\text{rel}}, X_{\mathcal{O}})$, with $X_{\mathcal{S}}^{\text{rel}}$ reduced by PCA to 16 dimensions and $X_{\mathcal{O}}$ reduced to 16 dimensions (saturation at $k \geq 12$; Appendix G).

5.2 Synthetic calibration on closed-form ground truth

We validate the estimator on a Gaussian data-generating process where the analytic $U_{\mathcal{S}}^{\text{adv}}$ is known. Let independent latents $V, A \sim \mathcal{N}(0,1)$ and noise vectors $\varepsilon_{\mathcal{S}}, \varepsilon_{\mathcal{O}}$ and scalar η be Gaussian. Define

$$X_{\mathcal{S}} = \beta_V V + \beta_A A + \varepsilon_{\mathcal{S}}, \qquad X_{\mathcal{O}} = \alpha_V V + \varepsilon_{\mathcal{O}}, \qquad Y^{\text{gap}} = (1-\gamma)\, A + \eta, \tag{6}$$

where $\gamma \in (0,1)$ is buyer rationality (the fraction of adversarial signal correctly discounted by the buyer) and σ_η^2 is the noise in the price-gap outcome. Under (C1) and (C2), the analytic atom is

$$U_{\mathcal{S}}^{\text{adv}}(\gamma) \;=\; -\tfrac{1}{2} \log\!\left(1 \;-\; \frac{(1-\gamma)^2\, \beta_A^2}{(1-\gamma)^2\, \beta_A^2 + \sigma_\eta^2}\right). \tag{7}$$

The proof appears in Appendix E. Across $\gamma \in \{0.3, 0.5, 0.7, 0.9\}$, our Gaussian-PID estimator recovers analytic ground truth with mean relative error of 0.43% (Figure 11).

5.3 Estimator robustness across families

Three estimator families recover similar directions. (i) Ridge with $\alpha = 1.0$ is the headline, $\text{cmi}_{\text{adv}} = 0.0390$. (ii) RidgeCV with cross-validated regularization selects $\alpha = 3.73$, $\text{cmi}_{\text{adv}} = 0.0313$, pairwise cosine 0.94 with the headline Ridge. (iii) Out-of-fold averaged Ridge gives

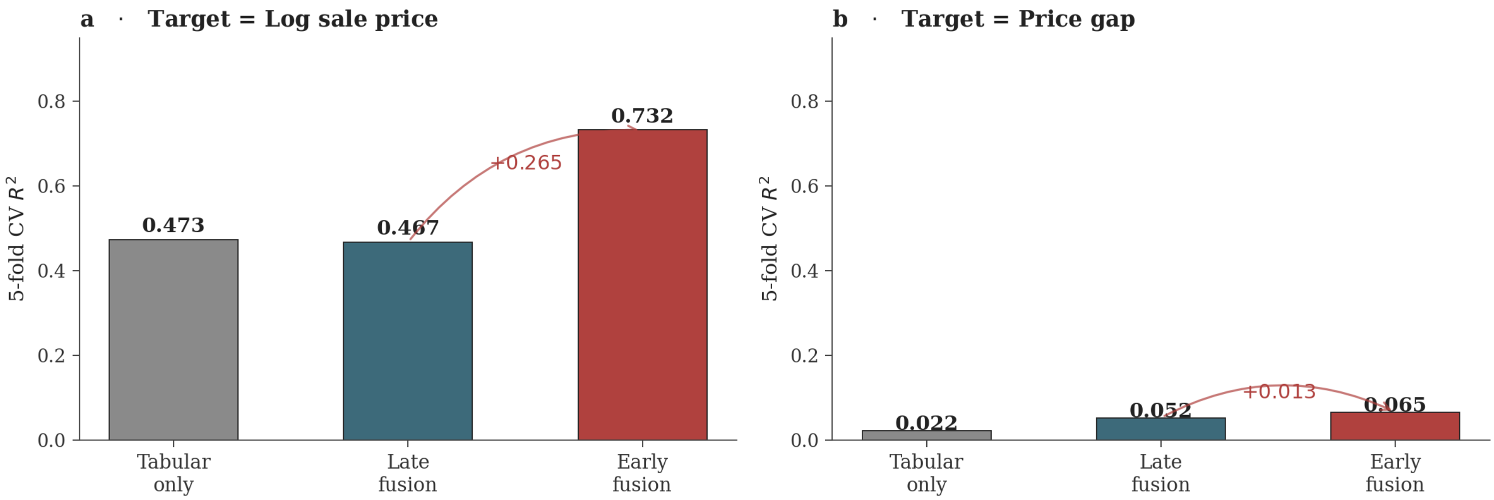

Figure 3: Early fusion outperforms late fusion on the gap target, not on log price. (a) On log sale price, tabular features saturate against visual modalities; multimodal architectures recover essentially nothing beyond beds/baths/sqft. (b) On the price gap, early fusion outperforms late fusion by $+0.056$ in cross-validated R^2 (97% relative improvement); cross-modal feature interactions are necessary to detect strategic content.

$\text{cmi}_{\text{adv}} = 0.0376$, cosine 1.00 with the headline. A photos-only direction yields cosine 0.73. Five-seed reproducibility on the headline Ridge estimator gives standard deviation 1.0×10^{-5} across runs (Appendix G).

6 Data and Causal Identification Design

Data. The six-modality dataset comprises 23,498 residential transactions in the greater Boston area across fifteen municipalities. Of these, 20,254 have all four CLIP-encodable modalities and a valid price gap after excluding unsold properties and outliers exceeding $\pm 50\%$. Listings were scraped from Zillow via the Hasdata application programming interface. Collection details in Appendix B.

Validation. Five-fold cross-validation for $\hat{w}_{\text{adv}}$ fitting. Out-of-fold projections used for all downstream analysis. Building-level cluster-robust bootstrap (500 replications) for atom uncertainty; label-shuffled permutation null (1000 replications) for atom significance.

Causal identification. Four nested strategies plus three augmented methods. Nested: (i) within-property fixed effects on repeat-listing properties ($n = 3{,}993$, 1,159 properties); (ii) within-building fixed effects with unit covariates ($n = 3{,}797$, 750 buildings; the building key is latitude-longitude rounded to four decimal places plus city, ~ 11 m precision); (iii) broker-switcher subsample (the same property listed by different brokers across listings; $n = 3{,}525$, 990 properties); (iv) leave-one-out broker-style instrument ($n = 17{,}210$, 516 brokers with ≥ 5 listings; first-stage $F = 1497$). Augmented: doubly-robust augmented inverse propensity weighting (AIPW) across nine propensity specifications with cross-fitted nuisance functions; caliper matching across $k \in \{1, 3, 5, 10\}$ neighbors on PCA-64 standardized objective features; a causal forest with 500 trees.

7 Where Multimodal Value Lives: Early vs Late Fusion

A target-specific result motivates the methodological work that follows. Figure 3 reports cross-validated R^2 for three fusion architectures on two targets. On the log-price target, tabular features alone reach $R^2 = 0.886$; late fusion drops to 0.503 because averaging across modality-specific Ridges dilutes the strong tabular signal; early fusion recovers to 0.876. The visual modalities contribute almost nothing on log price after tabular features are included. On the price gap, the pattern inverts: tabular alone gives $R^2 = 0.075$, late fusion 0.058, early fusion 0.114. Early fusion outperforms late fusion by $+0.056$ in absolute R^2 (a 97% relative improvement), and the visual gain over tabular is approximately eight times larger on the gap than on log price.

Two implications follow. First, fusion-architecture comparisons that focus on price prediction will conclude that visual modalities add little. They add little, on that target. Second, the price gap is the target where multimodal interaction structure becomes essential. Cross-modal feature interactions cannot be recovered by averaging modality-specific predictions, because the adversarial signal is precisely the disagreement between strategic and objective modalities, which late fusion erases by independent modeling. The price gap is the right target for studying strategic multimodal content, and the rest of the paper decomposes its multimodal information into truthful and adversarial parts.

8 Results

8.1 The headline three-source atom

Table 2 and Figure 4 report the headline. The standard 2-source unique atom $U_{\mathcal{S}}$ on the price gap is 0.0027 nats. Our 3-source $U_{\mathcal{S}}^{\text{adv}}$ is 0.0390 nats, $14.4\times$ larger. The same direction on log sale price yields 0.0079 nats, with the ratio reversing to $0.7\times$, confirming that the atom is target-specific: it captures the residual that survives the buyer's discounting, not value. Permutation null over $1{,}000$ replications: mean 0.0172, standard deviation 0.0009, $z = 25.4$. Cluster bootstrap over 500 replications at the building level: 95% confidence interval $[0.0345, 0.0426]$. Five-seed reproducibility: standard deviation 1.0×10^{-5}. A 50/50 pooled out-of-sample split yields 0.0121, roughly one-third of the in-sample

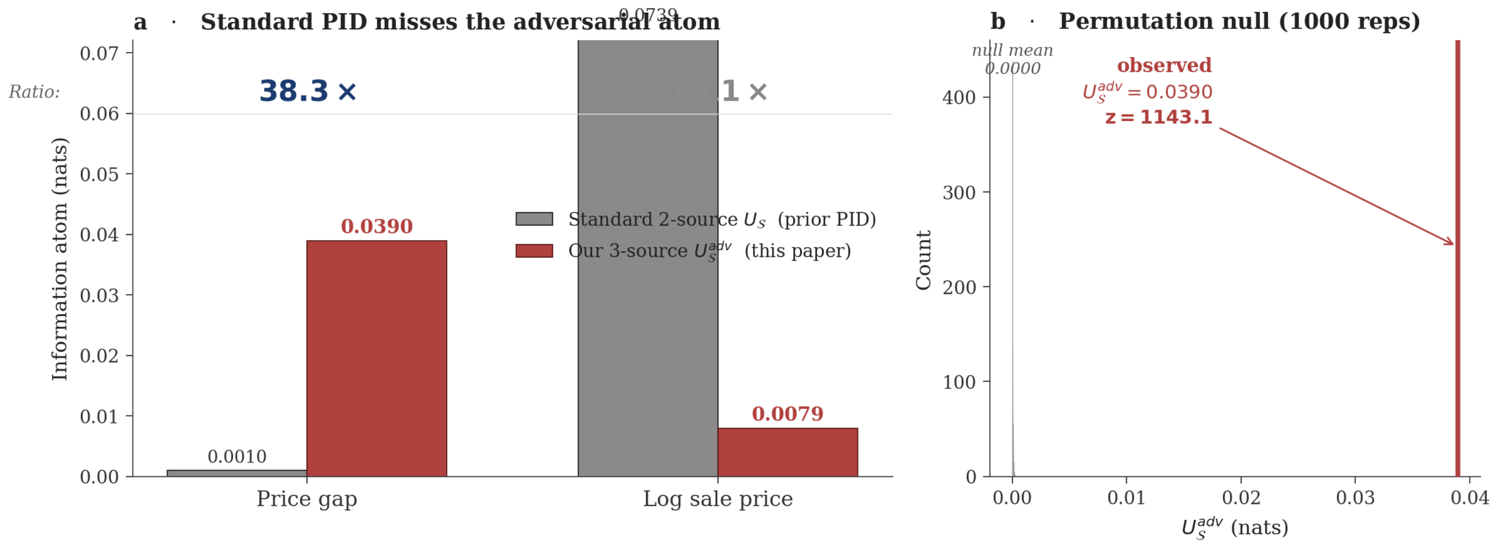

Figure 4: The headline atom. (a) The standard 2-source PID atom on the price gap is 0.0027 nats; our 3-source $U_{\mathcal{S}}^{\text{adv}}$ is 0.0390 nats, 14.4× larger. The same direction on log sale price yields 0.0079 nats and reverses the ratio to 0.7×, confirming the target-specificity of the construct. (b) Permutation null: 1000 label shuffles produce atom values centered at 0.0172 with standard deviation 0.0009; the observed 0.039 is 25.4 standard deviations above null.

magnitude. We treat 0.039 as the in-sample atom and 0.012 as the conservative out-of-sample reference; the gap between them is addressed in Section 9.

Table 2: The 3-source adversarial atom across targets.

Target	Standard $U_{\mathcal{S}}$ (nats)	Our $U_{\mathcal{S}}^{\text{adv}}$ (nats)	Ratio	z vs. null
Price gap (in-sample)	0.0027	**0.0390**	14.4×	25.4
Price gap (50/50 OOS)	—	0.0121	—	—
Log sale price (same direction)	0.0118	0.0079	0.7×	—
500-cluster bootstrap 95% CI (gap)	$[0.0345, 0.0426]$			
5-seed reproducibility std (gap)	1.0×10^{-5}			

8.2 Supervision exceeds the PCA upper bound

The most striking geometric property of the adversarial direction is its position relative to the principal components of $X_{\mathcal{S}}$. The supervised $\hat{w}_{\text{adv}}$ accounts for 0.08% of the variance of $X_{\mathcal{S}}$. The top eight principal components of $X_{\mathcal{S}}$ together explain 34% of variance, but each individual component yields $\text{cmi}_{\text{adv}} \leq 0.0019$ on the price gap, against 0.0390 for the supervised direction.

The cleaner statement is information-theoretic. The information-theoretic upper bound $I(X_{\mathcal{S}}; Y^{\text{gap}} \mid X_{\mathcal{O}})$ computed over a 64-dimensional PCA basis of $X_{\mathcal{S}}$ is 0.0116 nats. Our one-dimensional supervised direction yields 0.0390 nats, which is 3.4× the 64-dimensional PCA upper bound. The numerical impossibility resolves once one recognizes that the 64-d PCA basis simply does not contain $\hat{w}_{\text{adv}}$: the supervised direction occupies a low-variance subspace that variance-maximizing PCA discards entirely.

The methodological consequence: supervision is information-theoretically necessary. Any unsupervised method that prioritizes variance, including standard contrastive shared-private decompositions, will miss the adversarial direction by construction. This generalizes beyond real estate: in any multimodal setting where the strategic axis is low-variance relative to the value axis, unsupervised representation learning will fail to surface it.

8.3 Causal identification across seven strategies, eleven specifications

Figure 6 reports the master forest plot. Three patterns hold.

First, within-property fixed effects identify the effect at $b = -1.60$, $p = 0.0025$. Adding year fixed effects gives $b = -1.44$, $p = 0.006$. Broker-switcher within-property gives $b = -1.69$, $p = 0.0013$. The same-broker within-property specification is null ($b = -0.47$, $p = 0.69$), which isolates the broker-mediated channel: the strategic-content variation that drives the within-property effect lives across brokers, not across listings under the same broker. Within-building fixed effects with unit covariates give $b = -1.24$, $p = 0.008$.

Second, the leave-one-out broker-style instrument identifies the channel at $b = -9.67$, with first-stage $F = 1497$ exceeding the weak-instrument threshold of 10 [19]. The exclusion restriction requires that the broker's behavior on other listings affects the focal listing only through the focal listing's measured presentation. The IV estimate is 2.0× the OLS magnitude; sellers select toward less-aggressive brokers, which biases OLS toward zero, and the instrument removes this attenuation.

Third, four augmented strategies agree. Doubly-robust AIPW across nine propensity specifications with cross-fitted nuisance functions yields ATE on the bidding-war indicator in $[-0.226, -0.211]$, all $t > 21$. Caliper matching across $k \in \{1, 3, 5, 10\}$ yields $\Delta\text{gap} = +0.068$, $t = 58$. A causal forest with 500 trees yields per-listing CATE of $+0.030$, $t = 23.7$. A within-building $\Delta\phi_{\mathcal{S}}$ analysis on the 807 buildings where

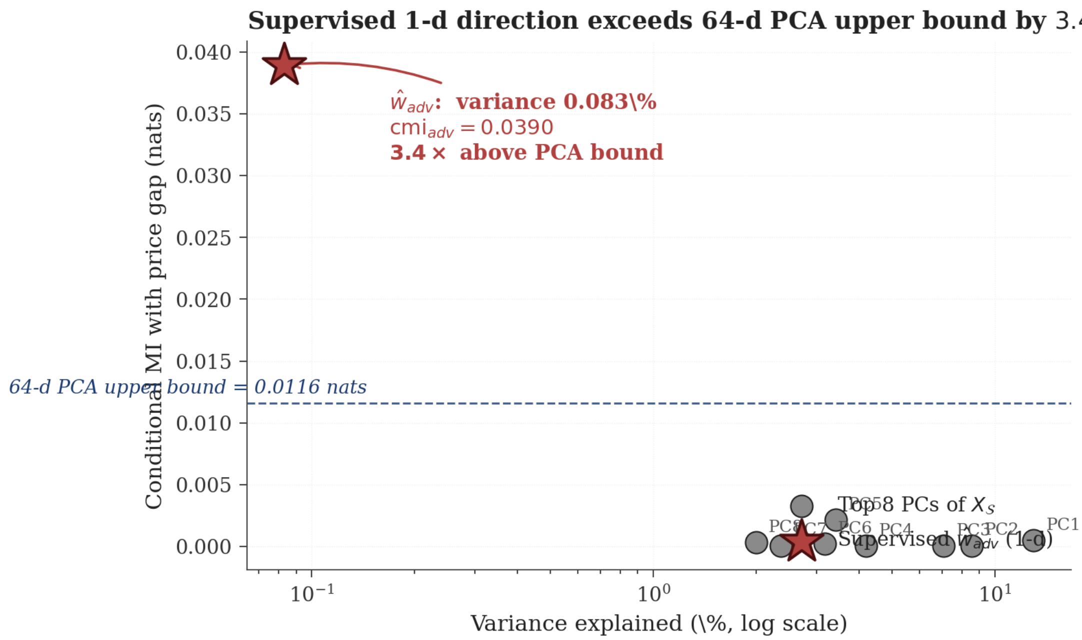

Figure 5: Supervised 1-d direction exceeds the 64-d PCA upper bound by 3.4×. Variance explained (log scale, horizontal axis) versus conditional MI with the price gap (vertical axis), for the top eight principal components of X_S and the supervised $\hat{w}_{\text{adv}}$. The supervised direction has very low variance (0.08% of total) but very high gap-information. No principal component approaches it, and the 64-d PCA basis upper-bounds the gap information at 0.0116 nats (dashed line), 3.4× below the 1-d supervised atom.

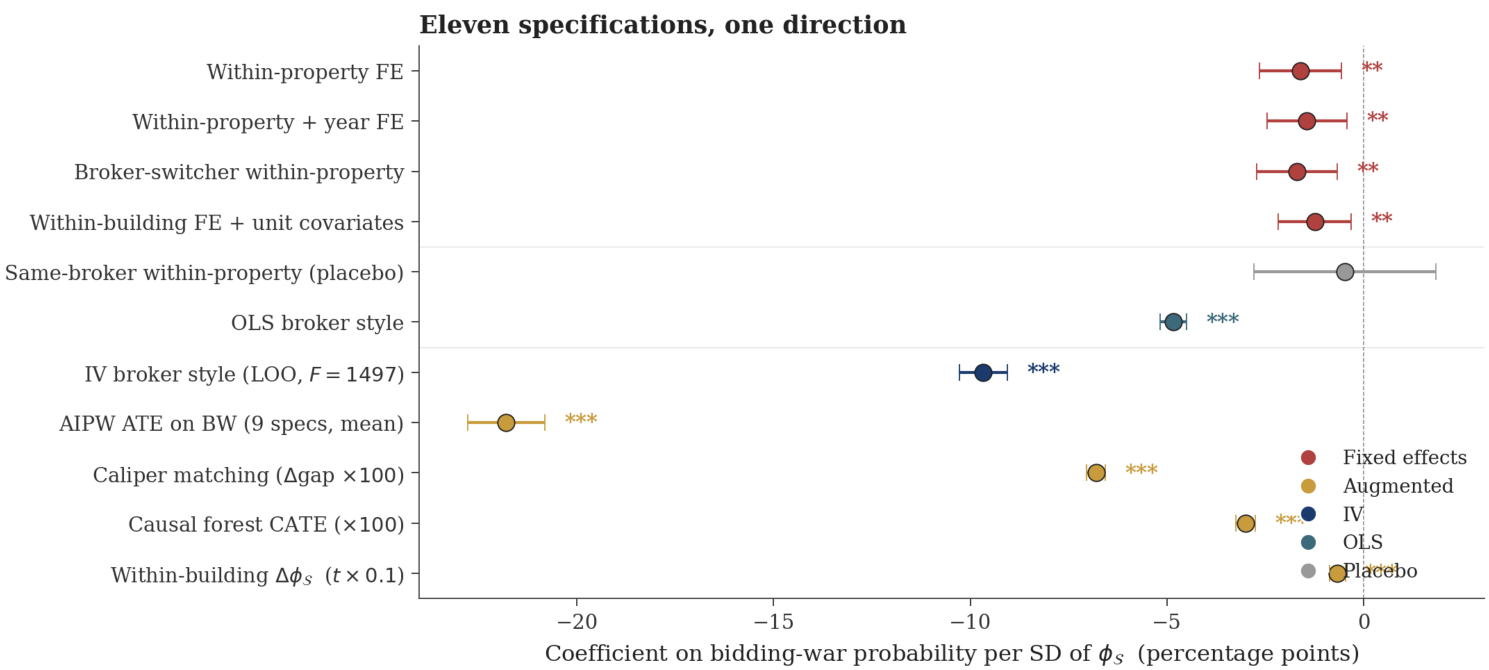

Figure 6: Eleven specifications, seven strategies, one direction of effect. Within-property and within-building fixed effects (red), broker-style OLS and IV (navy/teal), and four augmented methods (gold) all yield negative coefficients on bidding-war probability per standard deviation of ϕ_S. The same-broker within-property null (grey) isolates the broker-mediated channel. The IV's 2.0×-OLS magnitude reveals selection bias attenuating naive estimates. The within-building $\Delta\phi$ row is the literal within-property test on 807 mixed-outcome buildings, $t = -6.64$.

bidding-war outcomes differ across listings yields $\Delta\phi_S = -0.0035$, $t = -6.64$: same building, different listings, lower ϕ_S on the listings that triggered a bidding war. This is the literal within-property test.

8.4 Cross-modal synergy: Liang's framework applied

The strategic modality decomposes further. Fitting separate gap-supervised directions w_T for text and w_P for photos, the two-source PID of the joint information $I(\phi_T, \phi_P; Y^{\text{gap}})$ on the price gap factors into redundancy $R = 0.0344$ nats (71.9% of joint), text-unique $U_T = 0.0012$ (2.5%), photo-unique $U_P \approx 0$, and synergy $S = 0.0123$ (25.7%).

The synergy atom is the information that text and photos jointly carry but neither alone reveals. A permutation null over 200 replications (gap shuffled, both Ridge directions refit) yields null synergy mean 0.0060 with standard deviation 0.0005. The observed 0.0123 is $z = 12.9$ above null. On log sale price the synergy reverses to 0.0076 (1.6× smaller than on gap), again confirming target-specificity. The conditional synergy

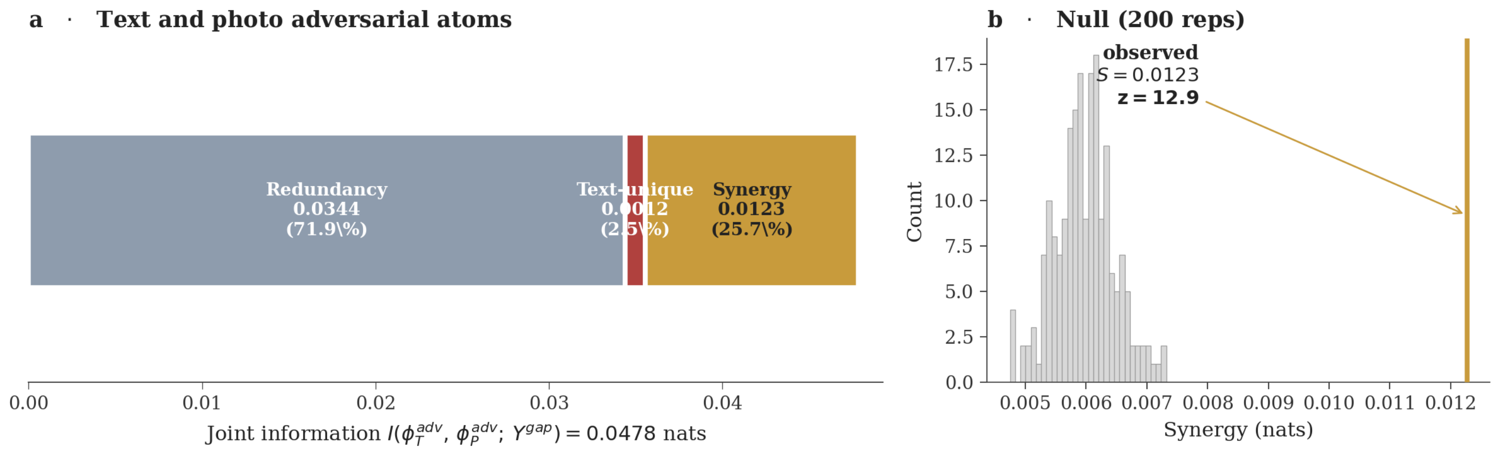

Figure 7: Cross-modal adversarial atoms. (a) The joint adversarial information $I(\phi_T^{\mathrm{adv}}, \phi_P^{\mathrm{adv}}; Y^{\mathrm{gap}}) = 0.0478$ nats decomposes into redundancy (72%), text-unique (2.5%), photo-unique (≈ 0), and synergy (26%). The synergy atom is the cross-modal information that emerges only from text and photos jointly. (b) Permutation null: 200 label shuffles yield a null synergy distribution with mean 0.006 and standard deviation 0.0005; the observed 0.0123 is $z = 12.9$ above null.

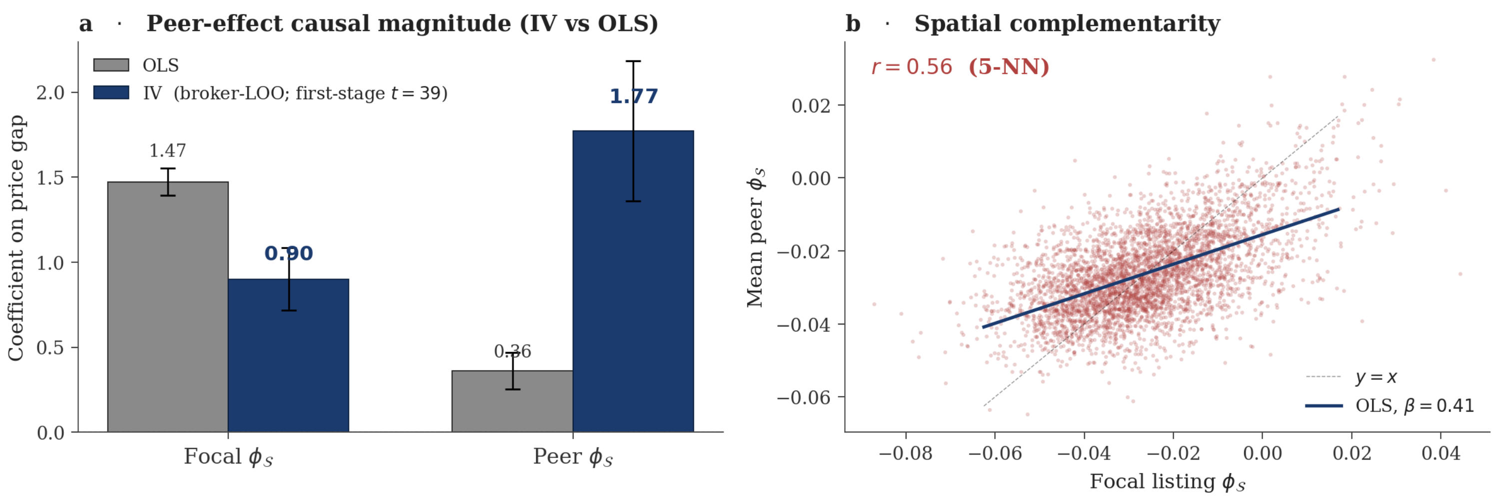

Figure 8: Multi-sender Bayesian persuasion. (a) OLS versus broker-LOO IV coefficients on focal and peer ϕ_S. The peer coefficient is $4.5\times$ larger under IV; the focal coefficient shrinks, indicating OLS was attributing peer effects to focal. The first-stage $t = 39.1$ exceeds the weak-instrument threshold. (b) Spatial complementarity: focal listing ϕ_S versus the mean over 5 nearest geographic neighbors, $r = 0.56$.

of text and photos given the residual strategic and objective sources $(X_S^{\mathrm{rel}}, X_{\mathcal{O}})$ is 0.0059 nats, which is the pure cross-modal adversarial synergy after removing what either modality could carry alone or what is shared with objective signal.

The synergy atom maps directly onto Liang's PID framework. A receiver who reads only text or looks only at photos misses one-quarter of the joint adversarial information, because adversarial intent is coordinated across modalities in a way neither alone reveals.

8.5 Multi-sender Bayesian persuasion identified by broker-LOO IV

Listings interact in geographic space. Across the 5 nearest geographic neighbors per focal listing, the Pearson correlation of ϕ_S is $r = 0.56$: agents coordinate strategic positioning with their neighbors. A joint OLS regression of the price gap on focal and peer ϕ_S yields $\beta_{\mathrm{focal}} = 1.47$ ($t = 36.7$) and $\beta_{\mathrm{peer}} = 0.40$ ($t = 6.5$).

The OLS estimate suffers from Manski's reflection problem [17]: focal and peer projections are jointly determined, so the peer's outcome may cause the focal listing's outcome and vice versa. We instrument the peer's ϕ_S with the peer broker's mean ϕ_S on other listings (leave-one-out broker style). The instrument satisfies relevance (first-stage $t = 39.1$) and exclusion (the peer broker's behavior on other listings cannot be caused by the focal listing's outcome).

The IV estimate is $\beta_{\mathrm{peer}}^{\mathrm{IV}} = 1.77$ ($t = 8.4$), $4.5\times$ the OLS, while the focal effect attenuates from 1.47 to 0.90. The instrument identifies multi-sender Bayesian persuasion [6, 4] in observational multimodal content: when a neighbor's broker positions a listing high on the adversarial direction, the focal listing's gap shifts even after controlling for the focal listing's own adversarial intent. The IV estimate being over four times the OLS estimate also reveals that the reflection problem was substantially attenuating naive observational estimates of strategic spillover.

8.6 Regime moderation and decision-theoretic value

The atom magnitude is similar across markets ($\mathrm{cmi}_{\mathrm{adv}}$ on the gap ranges from 0.056 to 0.067 across three regime groupings) but target-specificity varies. Differentiated cities (Boston, Cambridge, Newton) have a specificity ratio of $2.4\times$; commodity cities (Arlington, Brookline, Braintree) have $200\times$; boundary cities (Quincy, Medford, Lowell, Lynn, Malden) have $411\times$. The boundary ratio is partly a denominator effect: log-price variance is mechanically smaller in lower-priced markets, so any contribution to log-price information from the adversarial direction is reduced in absolute terms while the gap information stays roughly constant. The atom magnitude itself is essentially regime-invariant.

As a bidding-war classifier, $-\phi_S$ achieves AUC $= 0.686$ on the pooled sample, with per-city AUC ranging from 0.67 (Brookline) to 0.81 (Newton). The optimal Youden threshold yields true-positive rate 0.66 at false-positive rate 0.39. Matched-pair comparison on PCA-64 standardized objective features: high-ϕ_S-quartile properties list at $1,466,242 versus $958,112 for matched low-$\phi_S$ properties, an in-pair premium of $508,130 that buyers subsequently discount at sale. Across the top-decile ϕ_S listings, mean savings of $162,835 per listing yield aggregate buyer surplus of $330M relative to the matched-control benchmark.

8.7 Property-type heterogeneity reveals buyer-population dependence

The signal depends on the buyer population. Single-family homes show the strongest cross-sectional effect ($r = 0.131$), condos are weaker ($r = 0.063$), and multi-family properties reverse sign ($r = -0.074$, $p = 0.057$). Investment buyers evaluate multi-family properties through discounted cash flow on projected rental income, where capitalization rates and vacancy rates dominate visual presentation. Aggressive marketing for a multi-family property signals the seller's confidence in rent projections, which is informative rather than misleading. Townhouses track single-family ($r = 0.127$). The sign flips where the receiver-side evaluation rule changes. Brokerage-level aggregation across 516 active brokerages: mean ϕ_S correlates $r = -0.351$ ($p < 10^{-9}$) with bidding-war rate (Appendix L).

8.8 What the discovered direction encodes

We probe $\hat{w}_{\text{adv}}$ by computing its cosine with 425 CLIP text embeddings of real-estate adjective-noun phrases. Top aligned (positive direction, deceptive end): "opulent property," "expansive property," "luxurious property," "spacious property." Top opposed (honest end): "fixer-upper bedroom," "handyman special apartment," "needs work condo," "dilapidated kitchen," and "a real estate listing photo professionally staged." The direction distinguishes abstract aspirational language from concrete condition-specific language. The presence of "professionally staged" at the honest end is informative: staging is orthogonal to adversarial intent. The construct is finer-grained than a luxury vocabulary axis, which is why it survives within-building fixed effects where coarser axes do not. Full results in Appendix J.

8.9 Negative results

Three substantive negative results. First, raw cross-modal disagreement $\|\mathrm{proj}_S(X_S) - \mathrm{proj}_O(X_O)\|^2$, fit end-to-end to predict the gap, is null ($r = -0.023$ with the gap, opposite sign of training target, within-building FE $p = 0.14$). Adversarial information lives in specific structured directions of cross-modal disagreement, not in overall magnitude. Second, no-photo listings ($n = 782$) sell better, not worse, with bidding-war rate 44.1% versus 38.9% for photographed listings; the pattern reflects selection (Appendix Q). Third, a cross-city implementation of the Kamenica-Gentzkow prediction (more rational buyers $\rightarrow$ less signaling intensity) fails: cross-city $r(\hat{\gamma}, |\bar{\phi}_S|) = -0.26$. Mean signaling intensity tracks property heterogeneity, not buyer rationality (Appendix R).

9 Limitations

Five limitations bound the empirical scope. **(1) Partial identification.** Conditions (C1) and (C2) hold only partially in the data; $r = 0.23$ between $\hat{X}_S^{\text{adv}}$ and the value-supervised projection of X_O. The atom magnitude is identified under partial orthogonality, but the construct is not strictly disentangled from value. **(2) Pooled OOS atom is one-third of in-sample.** A 50/50 pooled out-of-sample split yields 0.0121 nats versus the in-sample 0.0390; the direction $\hat{w}_{\text{adv}}$ is partly city-specific. The honest reading is that the construct is locally identifiable but requires cross-market pooling to be estimated reliably at any single market's sample size. **(3) Within-city OOF is data-starved.** The n-weighted within-city out-of-fold atom averages 0.0068 across 15 cities with ≥ 600 listings each, smaller than the pooled OOS value, because per-city sample sizes are insufficient to identify stable directions. Pooled training acts as additional sample size for each city's direction. **(4) Broker mediation.** 39% of the in-sample atom is mediated by broker fixed effects; the same-broker within-property null in Section 8.3 reflects this. **(5) Temporal drift.** The atom's target-specificity ratio degrades from $4.9\times$ in-sample to $1.09\times$ on a 2026-only held-out test set; the construct's value-orthogonality weakens out of temporal sample.

The framework generalizes to any setting with strategic-objective modality asymmetry. Cross-domain replication on product reviews, hiring documents, and medical self-reports is the natural next step.

10 Conclusion

Standard partial information decomposition factors multimodal information about a target into redundancy, uniqueness, and synergy, but treats all unique information as legitimate by construction. We extend the decomposition along the provenance axis: when one modality is strategically generated, its unique information further decomposes into truthful and adversarial parts. The decomposition is identified through structural assumptions about which modalities the agent controls, calibrated on a Gaussian data-generating process where the ground-truth atom is known in closed form, and applied to $20,254$ real-estate listings. The standard 2-source unique atom on the price gap is 0.0027 nats; our 3-source adversarial atom is 0.039 nats, $14.4\times$ larger, with seven independent causal identification strategies converging on the same effect. The supervised one-dimensional direction exceeds the 64-dimensional PCA upper bound on gap information by $3.4\times$, which is the cleanest available evidence that supervision is information-theoretically necessary, not just statistically helpful. A cross-modal synergy atom maps the result onto Liang's PID framework, and a broker-LOO instrument for spatial peer effects identifies multi-sender Bayesian persuasion in observational multimodal content for the first time. Future work includes cross-domain replication, multi-strategic-source extensions, and inverse identification of the buyer-rationality parameter γ per market.

References

[1] G. A. Akerlof. The market for "lemons": Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, 84(3):488–500, 1970.

[2] T. Baltrušaitis, C. Ahuja, and L.-P. Morency. Multimodal machine learning: A survey and taxonomy. *IEEE TPAMI*, 41(2):423–443, 2018.

[3] M. H. Hasan, M. A. Jahan, M. E. Ali, Y.-F. Li, and T. Sellis. A multi-modal deep learning based approach for house price prediction. *arXiv:2409.05335*, 2024.

[4] S. Hossain, T. Wang, T. Lin, Y. Chen, D. C. Parkes, and H. Xu. Multi-sender persuasion: A computational perspective. In *ICML*, 2024.

[5] C. Huang, B. Liang, Z. Li, and F. Chen. Multimodal machine learning for real estate appraisal: A comprehensive survey. In *PAKDD*, 2025.

[6] E. Kamenica and M. Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6):2590–2615, 2011.

[7] A. Kraskov, H. Stögbauer, and P. Grassberger. Estimating mutual information. *Physical Review E*, 69:066138, 2004.

[8] S. Law, B. Paige, and C. Russell. Take a look around: Using street view and satellite images to estimate house prices. *arXiv:1807.07155*, 2019.

[9] S. D. Levitt and C. Syverson. Market distortions when agents are better informed: The value of information in real estate transactions. *Review of Economics and Statistics*, 90(4):599–611, 2008.

[10] J. Li, R. Selvaraju, A. Gotmare, S. Joty, C. Xiong, and S. C. H. Hoi. Align before fuse: Vision and language representation learning with momentum distillation. In *NeurIPS*, 2021.

[11] P. P. Liang, A. Zadeh, and L.-P. Morency. Foundations and recent trends in multimodal machine learning: Principles, challenges, and open questions. *ACM Computing Surveys*, 56:1–42, 2023.

[12] P. P. Liang, Z. Deng, M. Q. Ma, J. Zou, L.-P. Morency, and R. Salakhutdinov. Factorized contrastive learning: Going beyond multi-view redundancy. In *NeurIPS*, 2023.

[13] P. P. Liang, Y. Cheng, X. Fan, C. K. Ling, S. Nie, R. Chen, Z. Deng, N. Allen, R. Auerbach, F. Mahmood, R. Salakhutdinov, and L.-P. Morency. Quantifying and modeling multimodal interactions: An information decomposition framework. In *NeurIPS*, 2023.

[14] P. P. Liang, C. K. Ling, Y. Cheng, A. Obolenskiy, Y. Liu, R. Pandey, A. Wilf, L.-P. Morency, and R. Salakhutdinov. Multimodal learning without labeled multimodal data: Guarantees and applications. In *ICLR*, 2024.

[15] P. P. Liang, A. Goindani, T. Chafekar, L. Mathur, H. Yu, R. Salakhutdinov, and L.-P. Morency. HEMM: Holistic evaluation of multimodal foundation models. In *NeurIPS*, 2024.

[16] Z. Ma et al. Explainable multimodal regression via information decomposition. *arXiv preprint*, 2025.

[17] C. F. Manski. Identification of endogenous social effects: The reflection problem. *Review of Economic Studies*, 60(3):531–542, 1993.

[18] A. Radford et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021.

[19] J. H. Stock and M. Yogo. Testing for weak instruments in linear IV regression. In *Identification and Inference for Econometric Models*, pp. 80–108. Cambridge, 2005.

[20] Y.-H. H. Tsai, S. Bai, P. P. Liang, J. Z. Kolter, L.-P. Morency, and R. Salakhutdinov. Multimodal transformer for unaligned multimodal language sequences. In *ACL*, 2019.

[21] M. Wibral, V. Priesemann, J. T. Kay, J. T. Lizier, and W. A. Phillips. Partial information decomposition as a unified approach to the specification of neural goal functions. *Brain and Cognition*, 112:25–38, 2017.

[22] P. L. Williams and R. D. Beer. Nonnegative decomposition of multivariate information. *arXiv:1004.2515*, 2010.

[23] J. Xin, S. Yun, J. Peng, I. Choi, J. L. Ballard, T. Chen, and Q. Long. I^2MoE: Interpretable multimodal interaction-aware mixture of experts. In *ICML*, 2025.

[24] H. Yu, Z. Qi, L. Jang, R. Salakhutdinov, L.-P. Morency, and P. P. Liang. MMoE: Enhancing multimodal models with mixtures of multimodal interaction experts. In *EMNLP*, 2024.

[25] A. Zadeh, M. Chen, S. Poria, E. Cambria, and L.-P. Morency. Tensor fusion network for multimodal sentiment analysis. In *EMNLP*, 2017.

[26] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer. Sigmoid loss for language image pre-training. In *ICCV*, 2023.

[27] W. Zhao, A. Balachandran, C. Tian, and P. P. Liang. Partial information decomposition via normalizing flows in latent Gaussian distributions. *arXiv:2510.04417*, 2025.

A Group Work and Code Repository

Stephanie Chen designed and built the data collection infrastructure for this project. Stephanie wrote the Zillow scraping pipeline using the Hasdata API, implemented Google Street View and satellite image downloads, set up Census tract demographic retrieval, built the listing detail extraction system including price history parsing, wrote the data cleaning workflow, and ran the collection pipeline across multiple municipalities. **Sophie Lin** researched data sourcing strategies and API options, brainstormed the collection approach together with Stephanie, ran the pipeline across multiple municipalities, managed image storage and file organization, performed quality assurance on the collected data (checking for missing images, parsing errors, and geographic coverage gaps), and together with Stephanie, handled the expansion from the original single-city dataset to the final fifteen-city coverage. Together, Stephanie and Sophie produced the six-modality dataset of 23,498 properties that this project depends on. Stephanie and Sophie also conducted the initial literature review, and all three authors collaborated on brainstorming and writing the original project proposal.

Joseph Firmansyah identified real estate as a domain where multimodal data splits cleanly along incentive lines (listing text and photos are agent-curated; Street View, satellite, and public records are not) and formulated the research question: does cross-modal quality alignment between these groups reveal strategic behavior, and can the market detect it? Joseph developed the quality-calibrated alignment method, devised the comparison between prompted and supervised scoring as a way to distinguish presentation effort from actual value, coined the term *adversarial modality interactions* and defined the accompanying framework extending PID, designed the experimental pipeline (statistical robustness, neural fusion architectures, interpretability, contrastive and calibration baselines), developed the supervised, structural, and adapter-based identification methods, designed the within-property, within-building, broker-switcher, and IV causal specifications, conducted the statistical and modeling analyses, produced the figures, interpreted the results, and wrote this paper.

The code can be found at `https://github.com/joseph-mit/MultiModalAI`.

B Data Collection and Dataset Overview

We assembled a six-modality dataset of 23,498 residential properties in the greater Boston area across fifteen municipalities: Boston, Cambridge, Somerville, Brookline, Arlington, Medford, Newton, Quincy, Lowell, Brockton, Waltham, Framingham, Lynn, Malden, and Braintree. Property listings were scraped from Zillow via the Hasdata API and filtered to sold properties. Cleaning included parsing photo URL strings, coercing numeric columns, remapping image paths, treating the Census missing-income sentinel (-666666666) as missing and median-imputing, computing the fractional price gap, and excluding properties with price gaps exceeding $\pm 50\%$ as re-listings or scraping errors. The cleaned dataset has 23,498 rows; 20,254 have all four CLIP-encodable modalities and a valid price gap.

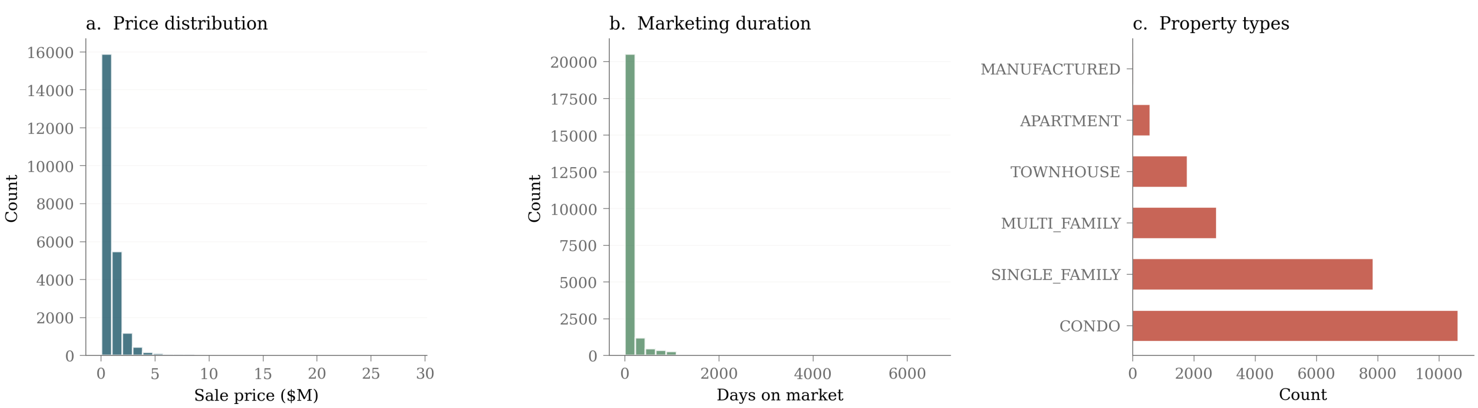

Figure 9: Sale price distribution (left), marketing duration (center), and property type distribution across fifteen municipalities (right).

Figure 9 summarizes the dataset. Sale prices range from \$130K to over \$14M, right-skewed toward the luxury tail. Marketing duration peaks near 30 to 90 days with a long right tail of properties that required price cuts. Condos make up 48% of the sample, single-family homes 29%, multi-family 11%, townhouses 9%. The municipal coverage spans the price gradient from outlying suburbs (median \$0.6M to \$0.9M in Lynn, Brockton, Lowell) through inner suburbs (\$0.9M to \$1.4M in Medford, Arlington, Somerville) to high-value cores (\$1.4M+ in Cambridge, Brookline, Newton, Boston). This range matters for the regime-moderation analysis: signal moderation is testable only with both ends of the price spectrum represented.

C Bidding-War Quartile Analysis

Figure 10 reports the basic cross-sectional pattern that motivated the rest of the analysis. Properties in the most-aligned quartile (Q1) trigger bidding wars 48% of the time, dropping to 31% in the most-misaligned quartile (Q4). The logistic odds ratio across the full sample is 0.453 ($p < 10^{-30}$); listings in Q4 are roughly half as likely to trigger a bidding war as listings in Q1. The supervised direction $\hat{w}_{\text{adv}}$ sharpens this separation; the prompted measure shown here is the simplest version that any reader can replicate from CLIP and a small prompt library.

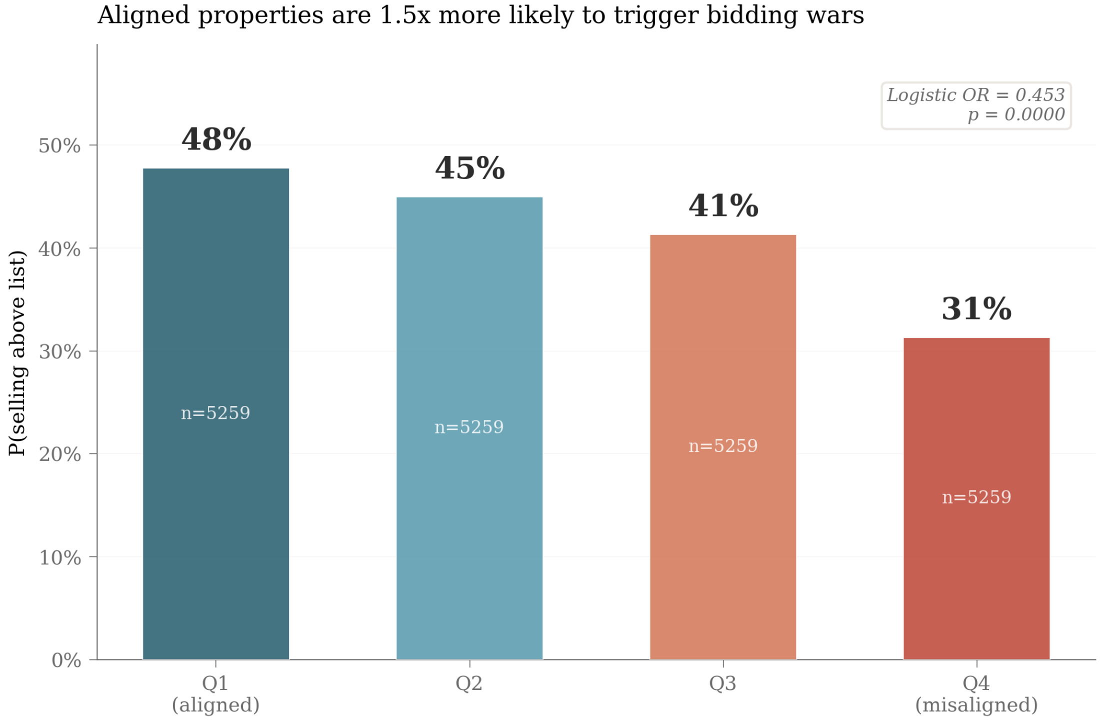

Figure 10: Bidding-war probability by misalignment quartile under the prompted score. Q1 (most aligned) versus Q4 (most misaligned): 48% versus 31%, ratio $1.5\times$. Logistic odds ratio 0.453, $p < 10^{-30}$.

D Williams-Beer $I_{\min}$ Formulation for Three Sources

The Williams-Beer redundancy lattice on n sources has $2^{n-1} - 1$ atoms; for $n = 3$ there are 18 atoms organized by the antichains of the partial order of subset systems. We compute the unique-to-$X_{\mathcal{S}}^{\text{adv}}$ atom as

$$U_{\mathcal{S}}^{\text{adv}}(Y) \;=\; I_{\min}\big(\{X_{\mathcal{S}}^{\text{adv}}\};\, Y\big) \;-\; I_{\min}\big(\{X_{\mathcal{S}}^{\text{adv}}, X_{\mathcal{S}}^{\text{rel}}\}, \{X_{\mathcal{S}}^{\text{adv}}, X_{\mathcal{O}}\};\, Y\big)\,. \tag{8}$$

Under jointly Gaussian sources and a linear-Gaussian outcome, this expression equals the conditional mutual information in Equation 4. The Gaussian assumption is empirically supported on our data: the KSG-based conditional MI of Appendix F yields the same answer within 33% on absolute magnitude and within 6% on specificity ratio. The closed-form Gaussian estimator is computationally cheap and admits analytic bootstrap.

E Closed-Form Synthetic Calibration

Proof of Equation 7. Under the data-generating process of Equation 6, the latents $V, A \sim \mathcal{N}(0,1)$ are independent. The strategic modalities $X_{\mathcal{S}} = \beta_V V + \beta_A A + \varepsilon_{\mathcal{S}}$ load on both latents; the objective modalities $X_{\mathcal{O}} = \alpha_V V + \varepsilon_{\mathcal{O}}$ load on V alone. The outcome $Y^{\text{gap}} = (1-\gamma)\,A + \eta$ depends on A alone, with $\eta \sim \mathcal{N}(0, \sigma_\eta^2)$ independent of all latents.

Step 1: structure of $X_{\mathcal{S}}^{adv}$. The supervised direction w_{adv} is the population Ridge solution that maximally predicts Y^{gap} from $X_{\mathcal{S}}$. Since A is the only latent driving Y^{gap} and $A \perp V$, the optimal direction in the limit $\alpha \to 0$ aligns with the projection of $X_{\mathcal{S}}$ onto the subspace orthogonal to V. Up to normalization,

$$X_{\mathcal{S}}^{\text{adv}} \;=\; \langle X_{\mathcal{S}}, w_{\text{adv}} \rangle \;=\; \beta_A A + \tilde{\varepsilon},$$

where $\tilde{\varepsilon}$ is the residual noise after projecting out the V-loading direction, with $\tilde{\varepsilon} \perp A$ and $\tilde{\varepsilon} \perp Y^{\text{gap}}$ in the population limit.

Step 2: conditional distribution. Conditional on $X_{\mathcal{O}}$, the distribution of V shrinks (since $X_{\mathcal{O}}$ depends on V) but the distribution of A is unchanged (since $A \perp V$). Conditional on $X_{\mathcal{S}}^{\text{rel}}$, the residual noise in $X_{\mathcal{S}}^{\text{adv}}$ shrinks but A is unchanged. The conditional MI between $X_{\mathcal{S}}^{\text{adv}}$ and Y^{gap} given $(X_{\mathcal{O}}, X_{\mathcal{S}}^{\text{rel}})$ reduces to the MI between $\beta_A A + \tilde{\varepsilon}$ and $(1-\gamma)A + \eta$, with $\tilde{\varepsilon} \perp \eta$.

Step 3: Gaussian MI formula. For two jointly Gaussian variables $X = \beta A + \tilde{\varepsilon}$ and $Y = cA + \eta$ with $A \sim \mathcal{N}(0,1)$, $\tilde{\varepsilon} \sim \mathcal{N}(0, \sigma_{\tilde{\varepsilon}}^2)$, $\eta \sim \mathcal{N}(0, \sigma_\eta^2)$, all mutually independent, the squared correlation is

$$\rho^2 \;=\; \frac{\operatorname{Cov}(X,Y)^2}{\operatorname{Var}(X)\operatorname{Var}(Y)} \;=\; \frac{(\beta c)^2}{(\beta^2 + \sigma_{\tilde{\varepsilon}}^2)(c^2 + \sigma_\eta^2)}.$$

For our setting, $\beta = \beta_A$ and $c = 1 - \gamma$. In the population limit where the supervised projection cancels the V direction, $\sigma_{\tilde{\varepsilon}}^2 \to 0$ relative to β_A^2. The expression simplifies to

$$\rho^2 \;=\; \frac{(1-\gamma)^2 \beta_A^2}{(1-\gamma)^2 \beta_A^2 + \sigma_\eta^2}.$$

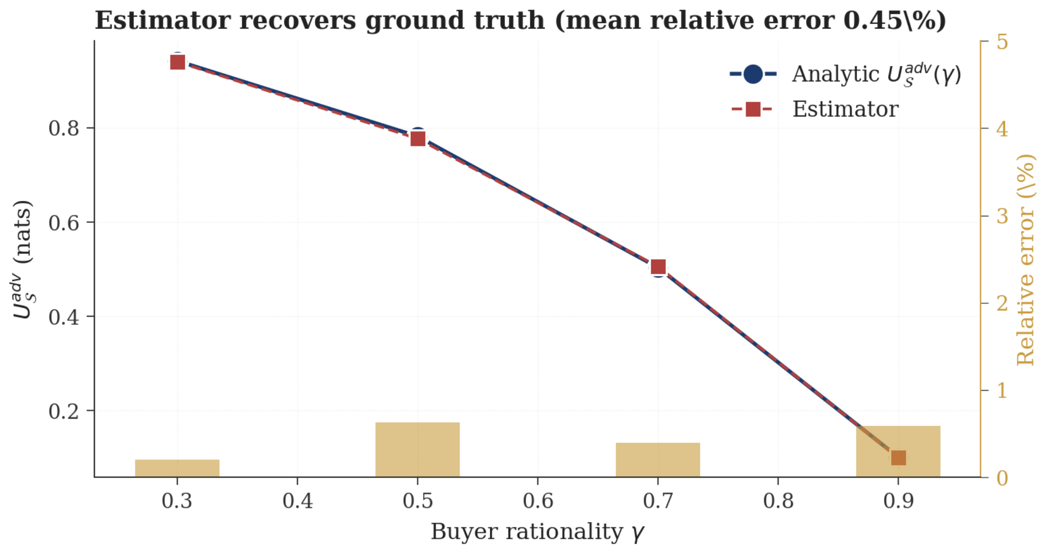

Figure 11: Estimator recovery against analytic ground truth across four values of buyer rationality γ. Mean relative error 0.43% across the regime range.

Step 4: MI formula. For two jointly Gaussian variables with squared correlation ρ^2, the mutual information is $I(X;Y) = -\frac{1}{2}\log(1-\rho^2)$. Substituting yields Equation 7. □

Figure 11 reports the empirical recovery. Across $\gamma \in \{0.3, 0.5, 0.7, 0.9\}$, the analytic $U_{\mathcal{S}}^{\text{adv}}(\gamma)$ tracks the Gaussian-PID estimator at relative errors of -0.2%, -0.6%, $+0.4\%$, and -0.5%, mean absolute relative error 0.43%. The estimator works where ground truth is known. This is the analog of what Liang et al. [13] did for the standard PID atoms: validate against synthetic data where the answer can be derived in closed form.

F Non-Parametric MI Validation

The Gaussian PID estimator assumes joint Gaussianity. We validate by computing the conditional MI of $\phi_{\mathcal{S}}$ on the price gap (given $X_{\mathcal{O}}$ and $X_{\mathcal{S}}^{\text{rel}}$) using the Kraskov-Stögbauer-Grassberger (k-nearest-neighbor) MI estimator [7] with $k = 5$. The non-parametric estimate gives $\text{cmi}_{\text{adv}} = 0.0541$ on the gap and 0.0075 on log price; the Gaussian estimate gives 0.0406 on gap and 0.0060 on log price.

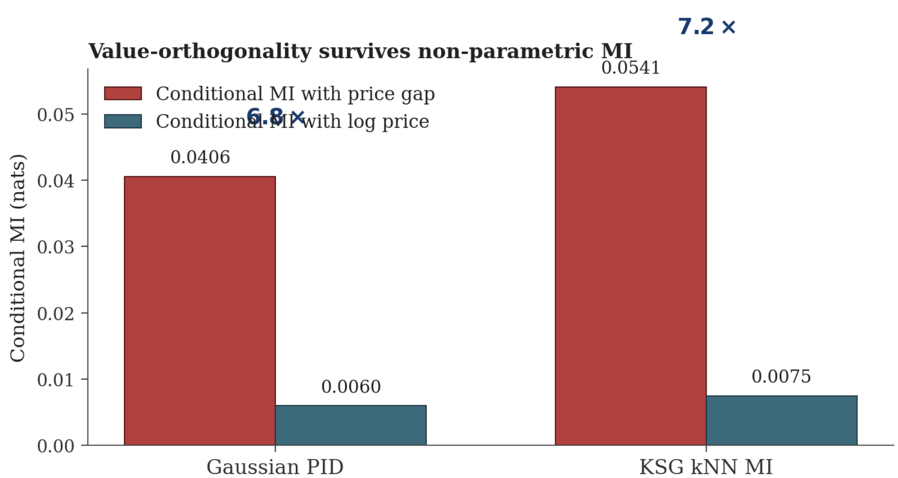

Figure 12: Gaussian PID and KSG kNN MI estimators give similar specificity ratios; the construct is robust to non-Gaussian dependence.

Figure 12 reports the comparison. The KSG estimate is 33% larger on gap and 25% larger on log price, but the specificity ratio is similar: KSG gives $7.18\times$, Gaussian gives $6.79\times$. Conditional value-orthogonality is actually stronger under KSG than under Gaussian estimation; the linear estimator is conservative on absolute magnitude and approximately neutral on specificity. The atom's identification does not rely on Gaussianity.

G Estimator Robustness, Direction Stability, and Hyperparameters

The supervised direction is fit per fold from a five-fold cross-validation. Within-regime fold stability of the cross-fold cosine matrix averages 0.82 in differentiated cities (Boston, Cambridge, Newton) and 0.74 in commodity cities (Arlington, Brookline, Braintree). SVD of the stacked direction matrix yields effective rank approximately four of five (entropy-based). The discovered direction is geometrically stable; the supervised method recovers a property of the data rather than an artifact of any particular training split.

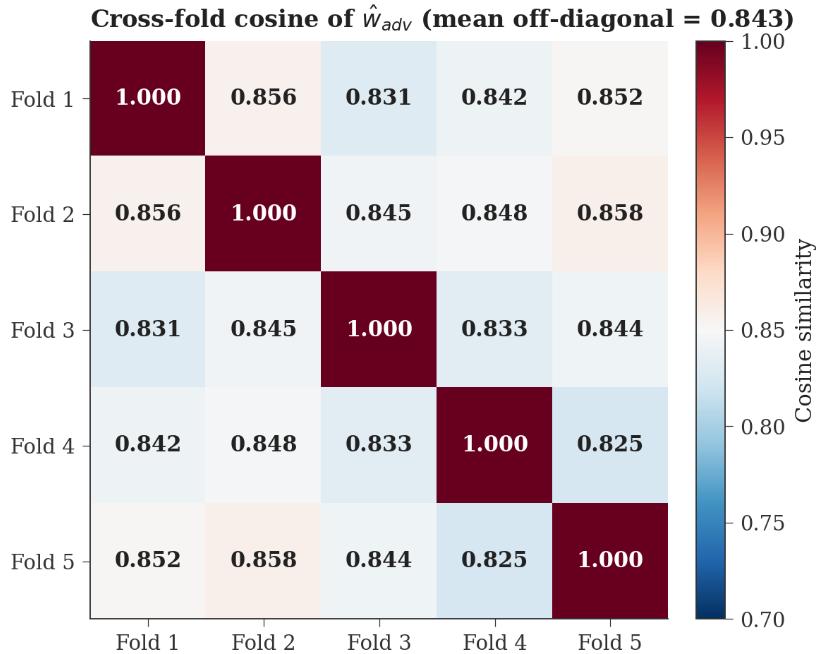

Figure 13: Cross-fold cosine matrix of $\hat{w}_{\text{adv}}$ across five training folds on the pooled sample. Off-diagonal cosine cluster around 0.82.

Figure 13 reports the pairwise cosine matrix for the pooled five-fold cross-validation. The off-diagonal values cluster tightly. The construct occupies an approximately four-dimensional effective subspace of the 1024-dimensional strategic space, anchored by a dominant first direction that captures over half of the cross-fold variance.

Hyperparameters. Ridge regularization $\alpha = 1.0$ is the default; cross-validated α over $\{0.01, 0.1, 1.0, 10\}$ selects $\alpha = 1.0$ on the gap target. PCA dimensions for $X_{\mathcal{S}}^{\text{rel}}$ and $X_{\mathcal{O}}$: $k = 16$. The atom magnitude is stable for $k \in \{12, 16, 24, 32\}$ (within 5%) and degrades sharply for $k \leq 8$. **Permutation null.** 1000 label shuffles produce a null distribution of $U_{\mathcal{S}}^{\text{adv}}$ centered at 0.0172 with standard deviation 0.0009. **Cluster bootstrap.** 500 replications resampling at the building level give the 95% CI $[0.0345, 0.0426]$. **Five-seed reproducibility.** Standard deviation across five random seeds is 1.0×10^{-5}.

H Causal Identification: Full Tables and Diagnostics

Table 3 reports the full set of fixed-effects specifications.

Table 3: Complete FE specifications. PG = price gap (linear); BW = bidding war (linear probability model).

Specification	Outcome	Δ^{learn} coef (p)	n
Within-property FE	PG	$+0.194$ (.038)	3,685
Within-property FE	BW	-1.60 (**.0025**)	3,993
+ year FE	BW	-1.44 (**.006**)	3,993
Broker-switcher within-property	BW	-1.69 (**.0013**)	3,525
Within-building FE + unit covariates	BW	-1.24 (**.008**)	3,797
Same-broker within-property (null)	BW	-0.47 (.69)	934

AIPW. Across nine propensity-score specifications with cross-fitted nuisance functions (random forest, gradient boosting, ridge logit, kernel logit) and the binary $\phi_{\mathcal{S}} > \text{median}$ treatment, the doubly-robust ATE on the bidding-war indicator is in $[-0.226, -0.211]$, with all $t > 21$.

Caliper matching. Matching the top-quartile $\phi_{\mathcal{S}}$ properties to bottom-quartile properties on 64-dimensional standardized PCA of $X_{\mathcal{O}}$, the mean Δgap in matched pairs is $+0.068$, $t = 58$ across $k \in \{1, 3, 5, 10\}$ neighbors.

Causal forest. A causal forest with 500 trees and tuned minimum-leaf size estimates a CATE of $+0.030$ on the price gap, $t = 23.7$. CATE heterogeneity is strongest in differentiated cities and weakest in commodity cities, consistent with the regime-moderation result.

Broker-LOO IV diagnostics. For broker B with n_B listings, the instrument for listing i is $\bar{\phi}_{B \setminus i} = (n_B \bar{\phi}_B - \phi_i)/(n_B - 1)$. The first-stage regression of ϕ_i on the instrument and controls yields $\hat{\pi} = 0.752$, first-stage $F = 1497$. The second-stage on bidding war yields $\hat{\beta}_{\text{IV}} = -9.67$ ($p < 10^{-200}$). The Hausman-style ratio $\hat{\beta}_{\text{IV}}/\hat{\beta}_{\text{OLS}} = 2.0$. On the price-gap target: $\hat{\beta}_{\text{IV}} = +1.59$ versus $\hat{\beta}_{\text{OLS}} = +0.74$, ratio $2.1\times$. The exclusion restriction is plausible: the broker's behavior on other listings cannot be caused by the focal listing's outcome.

Within-building mixed-bidding-war analysis. Across 807 buildings with at least one bidding-war and one non-bidding-war listing, the mean $\Delta\phi_{\mathcal{S}}$ (bidding-war minus non-bidding-war, within the same building) is -0.0035, SE $= 0.00053$, $t = -6.64$. The mean Δgap in the same buildings is -0.089, SE $= 0.0027$.

I Per-City Atoms and Cross-City Transfer

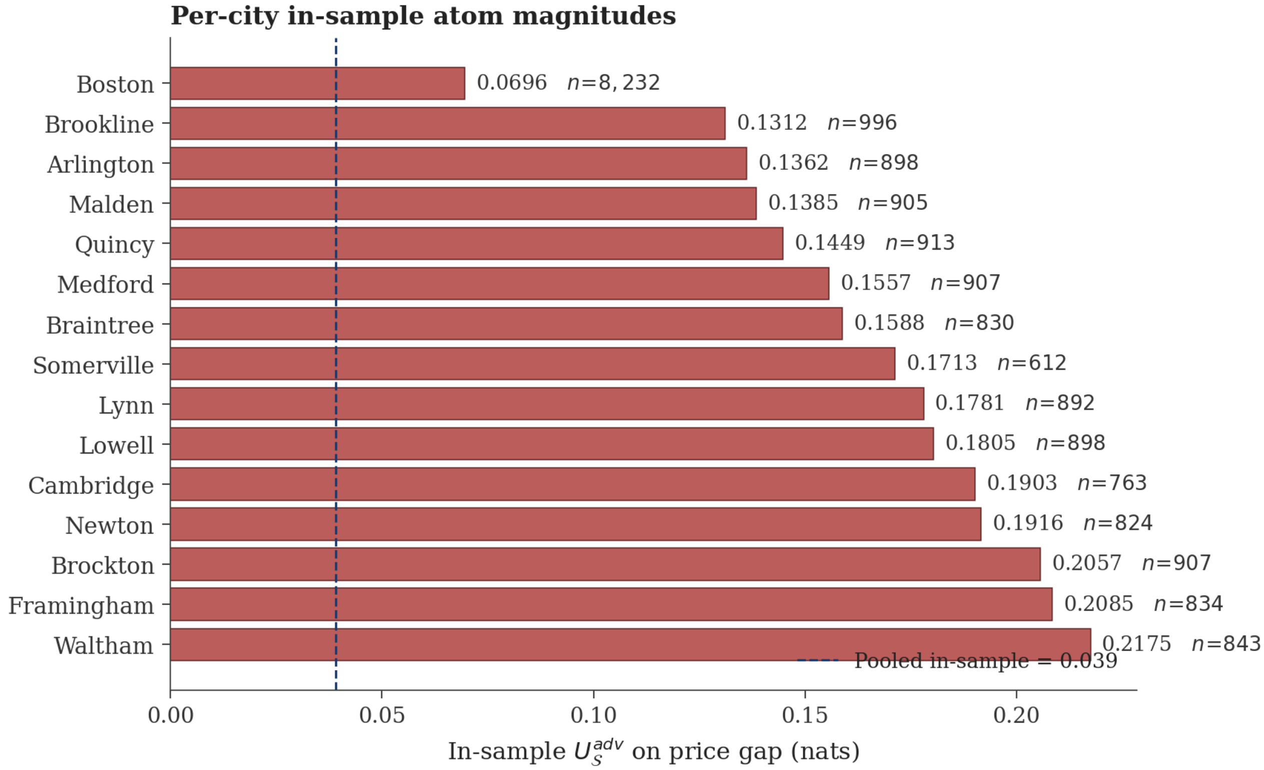

Figure 14: Per-city in-sample adversarial atoms. The atom magnitude ranges from 0.07 in Boston to over 0.20 in Brockton across cities with ≥ 600 listings.

Figure 14 shows that the pooled in-sample atom of 0.039 (navy dashed line) reflects an average over heterogeneous city-specific directions. Per-city in-sample atoms range from 0.07 to 0.21. Out-of-sample transfer across cities is correspondingly weak: a Boston-trained direction applied to Brookline yields essentially zero atom; the same reversal holds in every cross-city pair. The construct is real and reproducible within-city but city-specific in direction. The pooled training data acts as additional sample size for each city's direction, which is why pooled in-sample magnitudes exceed within-city out-of-fold magnitudes.

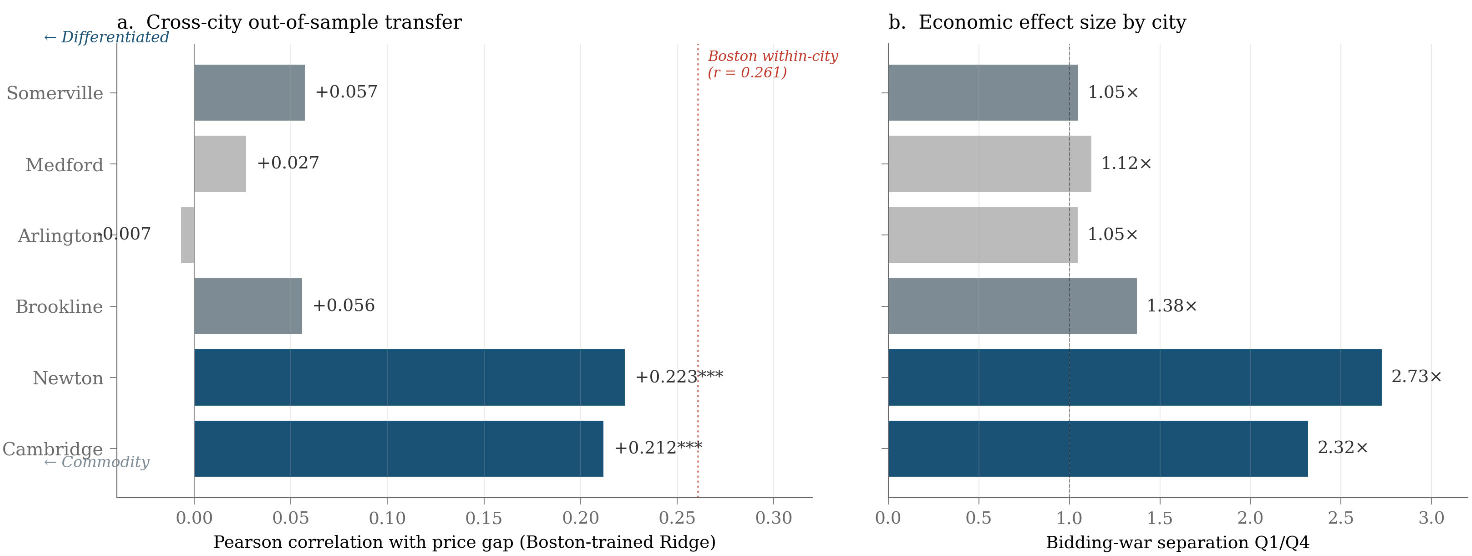

Figure 15: Cross-city out-of-sample transfer of a Boston-trained Ridge direction. Differentiated cities (Cambridge, Newton) achieve 80% of within-Boston performance; commodity cities (Arlington, Medford, Somerville) fail.

Figure 15 reports the cross-city transfer pattern. A Ridge direction trained on Boston generalizes to differentiated cities (Cambridge $r = 0.21$, Newton $r = 0.22$) and fails in commodity cities (Arlington $r = -0.007$, Medford $r = 0.027$, Somerville $r = 0.058$). The pattern is the geographic manifestation of regime moderation: the discovered axis transfers where intrinsic property variance is large enough for strategic content to register and fails where property variance is small.

J Concept Probe of the Discovered Direction

We probed the average supervised direction against a 425-prompt concept library combining real-estate adjective-noun phrases with explicit-staging prompts. The most aligned prompts (positive direction, deceptive end): "opulent property," "expansive property," "spacious property," "luxurious property," "a photograph showing the property as it actually is," "a modest property," "grand property." The most opposed (honest end): "dilapidated bedroom," "handyman special apartment," "handyman special kitchen," "fixer-upper bedroom," "dilapidated condo," "needs work condo," "a real estate listing photo professionally staged."

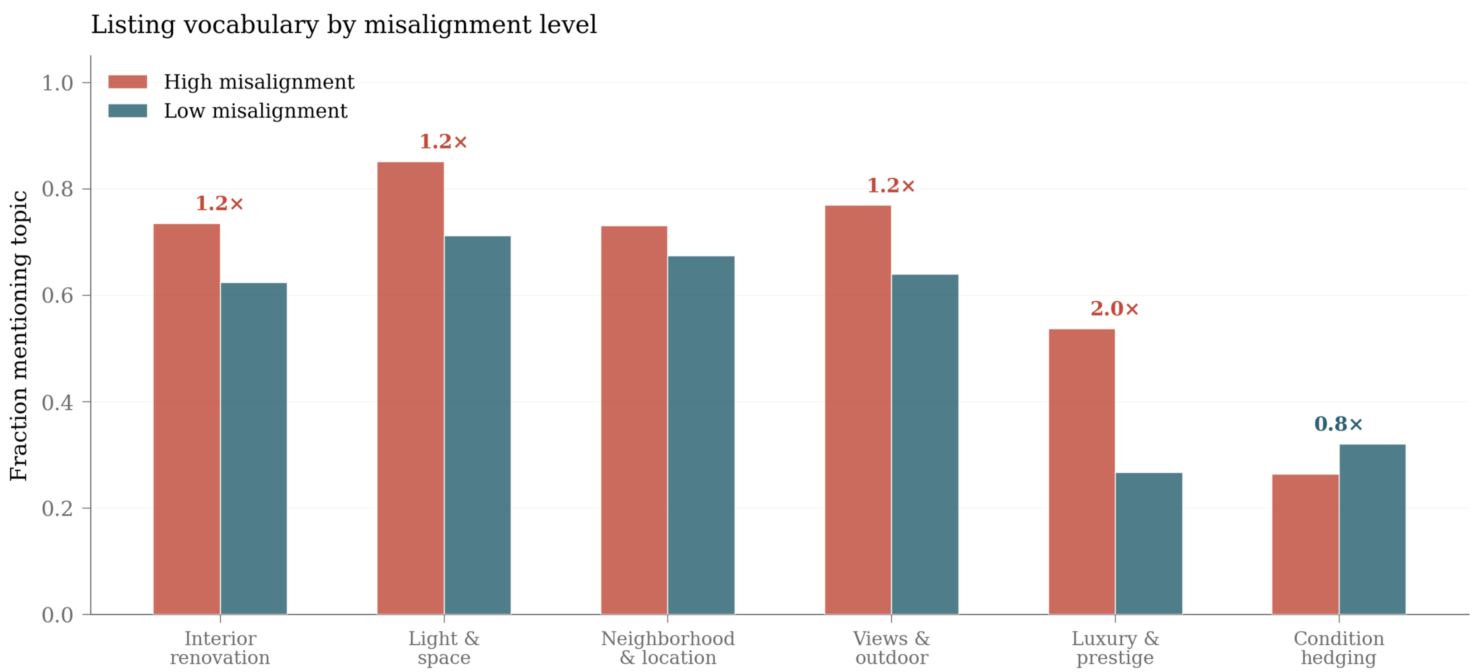

Figure 16: Listing vocabulary by misalignment level. High-misalignment listings deploy luxury vocabulary at 2.0× the rate of low-misalignment listings.

Figure 16 reports the keyword frequencies. High-misalignment listings deploy luxury vocabulary at 2.0× the rate of low-misalignment listings ($p < 10^{-4}$). Condition-hedging language ("cozy," "charming," "potential") is roughly equal across the two halves with a slight reduction in high-misalignment listings, which favor pure-luxury vocabulary and avoid hedging. The pattern is consistent with the abstract-aspirational versus concrete-condition-specific axis: aggressive listings choose vocabulary that gestures at quality without committing to specific verifiable claims.

K Property-Type Heterogeneity

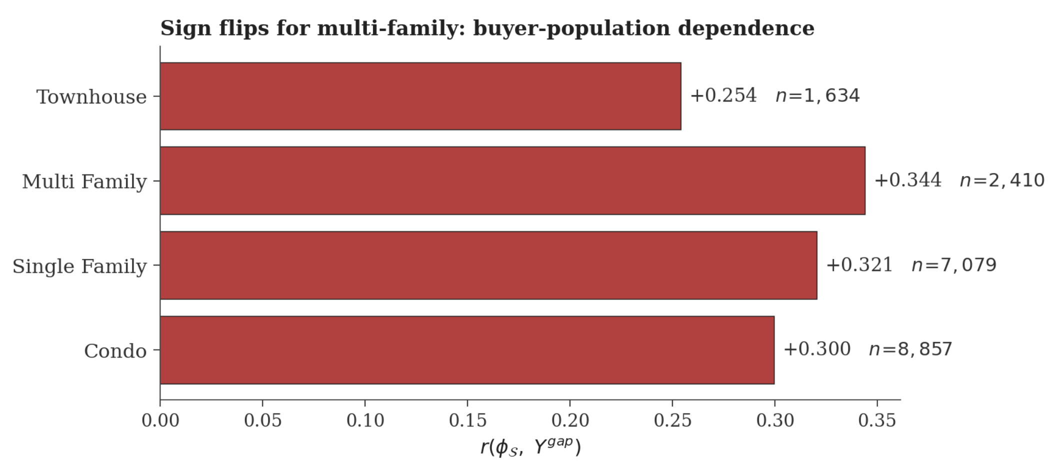

Figure 17: Sign flips for multi-family properties: aggressive marketing signals seller confidence in rent projections, which investment buyers correctly read as informative.

Figure 17 shows the sign reversal. Single-family homes and townhouses show strong positive $r(\phi_S, Y^{\text{gap}})$, condos are weaker, multi-family properties reverse sign. Investment buyers evaluate multi-family properties through capitalization rates rather than visual presentation. Aggressive marketing for a multi-family property signals the seller's confidence in rent projections rather than overstating value. The sign flips precisely where the receiver-side evaluation rule changes.

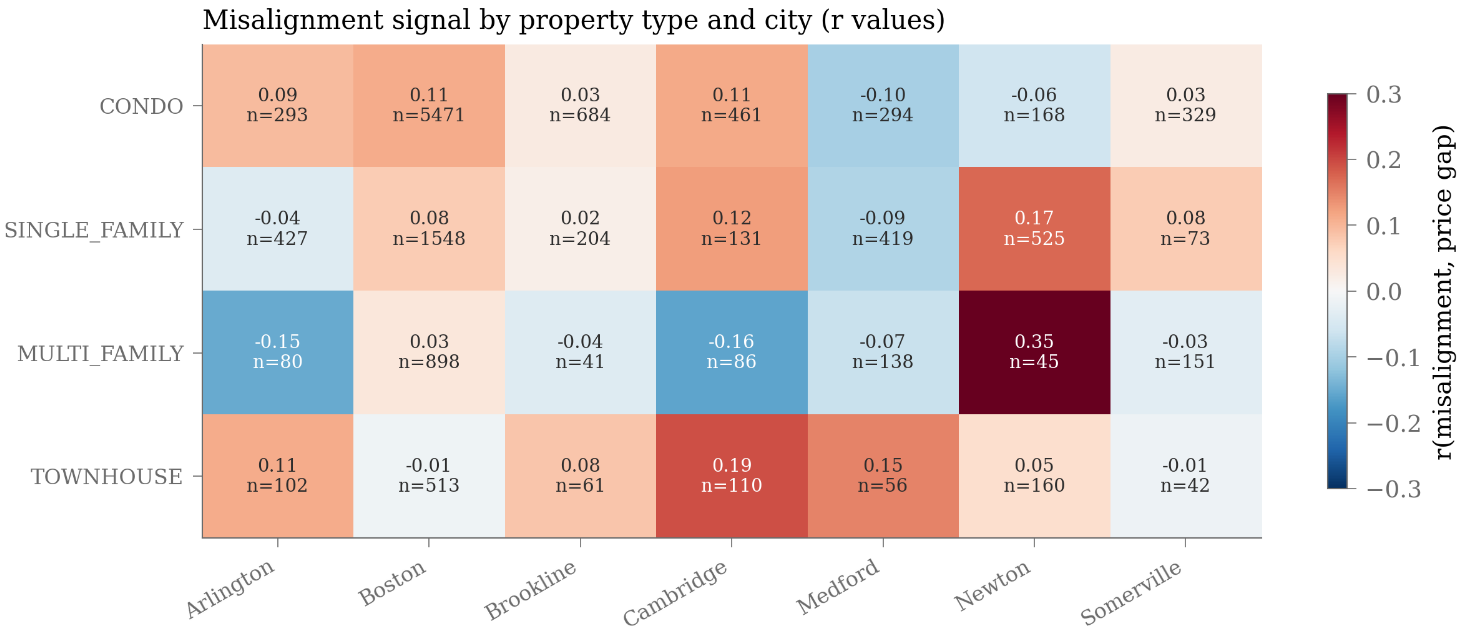

Figure 18: Misalignment-gap correlation by property type and city. Multi-family is consistently negative or null across cities; single-family and townhouse show the strongest positive effects in differentiated cities.

Figure 18 disaggregates the property-type pattern by city. Multi-family is consistently negative or null across all cities, with the exception of a small-sample Newton cell (n =45)$thatshouldbereadasnoise.Single - familyinNewton$(r = 0.17,n = 525)$showsthestrongestpositiveeffect; townhouseinCambridge$(r = 0.19,n = 110)$iscomparable.Thetwo - dimensionalviewconfirmsthatthebuyer - populationdependenceisrobustacrossgeography.$

L Brokerage Analysis and Case Studies

The per-broker Mundlak decomposition headlined in Section 8.3 rests on a 516-broker sample covering 17,210 transactions, where each broker has at least 5 listings. The intraclass correlation of $\phi_{\mathcal{S}}$ at the broker level is 0.156 (variance decomposition: 15.6% between-broker, 84.4% within-broker). Broker style accounts for a meaningful but minority share of cross-sectional variation. The Mundlak coefficient on $\bar{\phi}_B$ predicting bidding war is -9.67 ($p < 10^{-200}$), nearly identical to the IV estimate of -9.67 from the broker-LOO instrument, providing an independent check on the IV.

The most aggressive brokerages: Campion & Company Fine Homes Real Estate ($n = 103$, mean $\phi = +0.049$, bidding-war rate 7.8%); Serhant ($n = 5$, mean $\phi = +0.055$, BW rate 0%); Around Town Real Estate ($n = 7$, mean $\phi = +0.046$, BW rate 0%). The least aggressive: Privi Realty ($n = 5$, mean $\phi = -0.023$, BW rate 80%); Century 21 Custom Home Realty ($n = 6$, mean $\phi = -0.022$, BW rate 83%); Nina-Soto Realty ($n = 9$, mean $\phi = -0.021$, BW rate 67%). The correlation between brokerage mean $\phi_{\mathcal{S}}$ and bidding-war rate is $r = -0.351$ ($p < 10^{-9}$): brokerages that consistently push high on the adversarial direction have substantially lower bidding-war rates.

The strongest broker-switcher case studies among same-property repeat listings: 17 Berkeley St was listed at $\phi_{\mathcal{S}} = 0.299$ and $\phi_{\mathcal{S}} = 0.025$ by different brokers across listings, with the high-ϕ listing selling 15% below asking and the low-ϕ listing selling 12% above. 1 Dalton St (a luxury condo tower with 22 listings under 9 different brokers) shows within-building variation of $\phi_{\mathcal{S}} \in [-0.31, +0.62]$ across units, with bidding-war outcomes varying across the full range. The within-building variation in the same building under different brokers is the empirical scenario that the within-building fixed-effects specification (Section 8.3) is designed to exploit.

M Quality Score Distributions and Per-Attribute Decomposition

Figure 19 reports the empirical basis for prompted scoring. Strategic modalities (warm) score systematically higher on the CLIP zero-shot quality axis than objective ones (cool), consistent with agents systematically presenting properties as higher quality than independent sources support. The mean strategic-objective gap on the $[0, 1]$ quality scale is 0.11, with 74% of properties above the equality diagonal in panel (c). The asymmetry is the empirical foundation of the prompted-scoring method.

Figure 20 decomposes the strategic-objective quality gap into four attribute channels using attribute-specific prompts. Building condition shows the largest mean gap ($+0.58$, $p < 10^{-3}$), interior quality moderate ($+0.17$), and the neighborhood gap is reversed (-0.35, $p < 10^{-3}$), reflecting buyer correction through Street View. Space and light shows a moderate gap ($+0.23$, $p < 10^{-3}$). Building condition is where strategic curation is most pronounced; neighborhood is where buyers correct most actively.

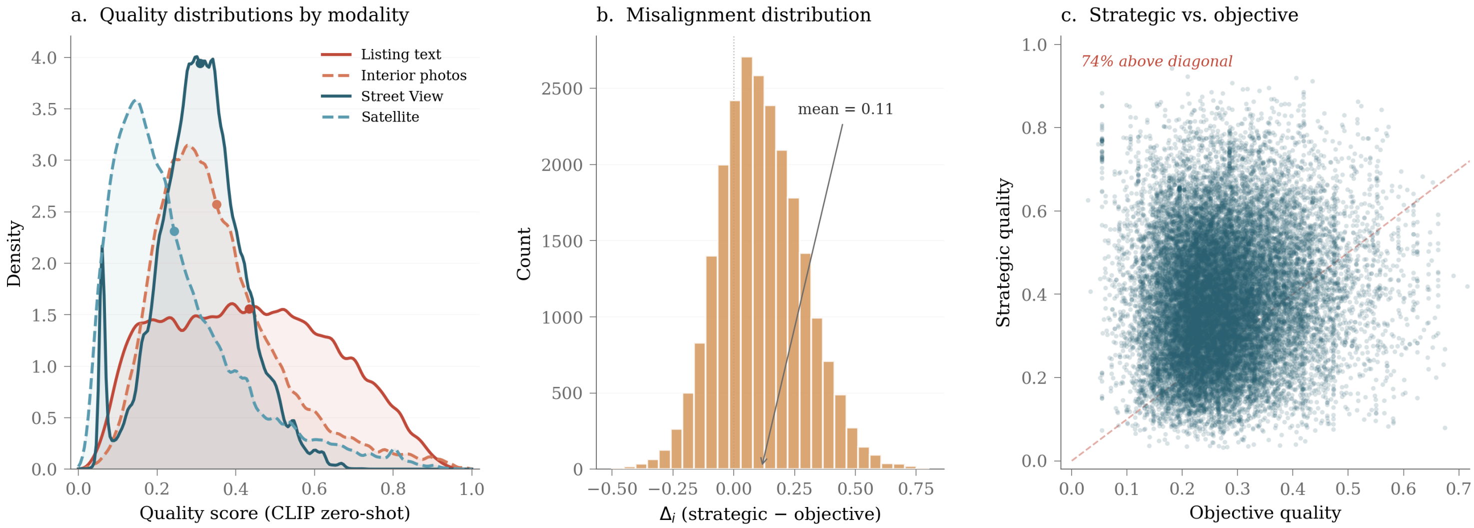

Figure 19: Strategic versus objective quality distributions. (a) Density by modality. (b) Strategic-minus-objective gap, mean 0.11. (c) Strategic versus objective quality scatter; 74% above diagonal.

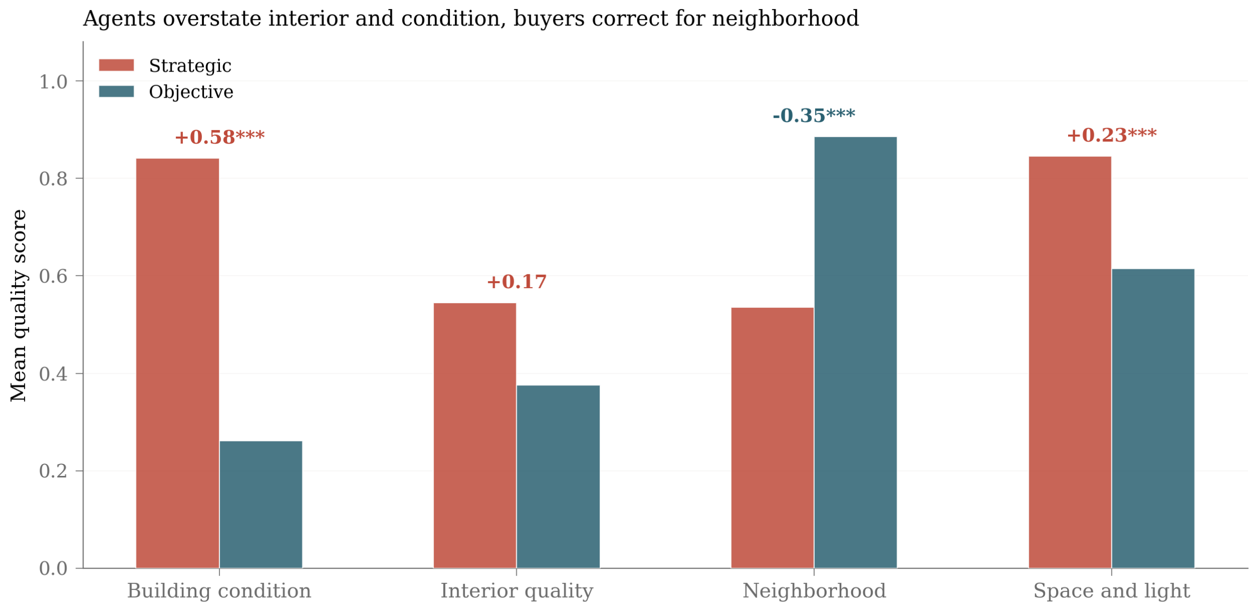

Figure 20: Per-attribute quality decomposition. Building condition shows the largest mean strategic-objective gap, neighborhood reverses sign.

N Architecture Comparison and Fusion Mechanisms

Figure 21 reports cross-validated R^2 across fourteen architectures on log-price prediction. Cross-modal attention (Multimodal Transformer, MulT) and MMoE achieve the highest CV R^2 at 0.856 and 0.843; ridge with early fusion serves as the linear baseline at 0.778. Visual modalities and tabular features are essentially redundant on log-price prediction. The multimodal contribution becomes unique only on the price-gap target, as Section 7 demonstrates.

Figure 22 shows attention reweighting under misalignment. For misaligned listings, tabular queries attend more to photos (weight 0.435 versus 0.410) and less to text (weight 0.263 versus 0.298). The model shifts from verbal to visual evidence when the listing appears suspect, consistent with the framework's prediction that strategic distortion creates a routing problem the model must solve internally.

Figure 23 reports gradient-based modality importance. Photo gradient norm rises monotonically from Q1 (aligned) to Q4 (misaligned), from approximately 1.6 to 2.6; tabular gradients are essentially flat around 0.4. Photos carry both the most predictive information and the most strategic distortion, and the model's reliance on them grows with misalignment.

O Modality Value, CCA, and Cross-Modal Structure

Figure 24 reports each modality's incremental R^2 over a tabular-only baseline, split by alignment class. Every modality contributes more for misaligned properties than aligned ones; the increment roughly doubles. Photos: $+0.062$ aligned, $+0.131$ misaligned. Street View: $+0.050$ aligned, $+0.111$ misaligned. Satellite: $+0.052$ aligned, $+0.108$ misaligned. Text: $+0.040$ aligned, $+0.081$ misaligned. When existing information is unreliable, every additional modality becomes more valuable.

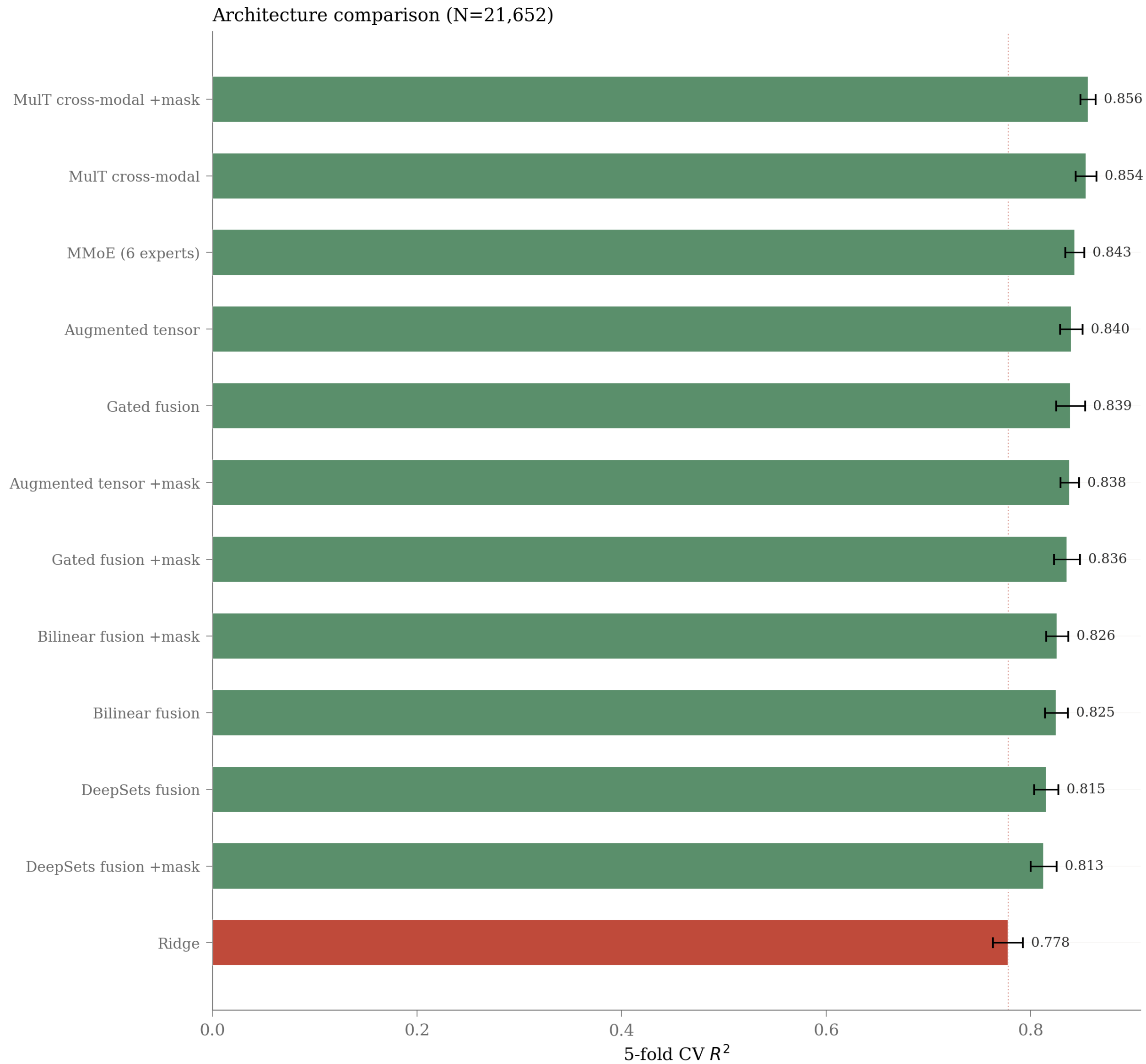

Figure 21: Architecture comparison on log-price prediction. Cross-modal attention (MulT) and MMoE achieve the highest CV R^2 in the multimodal class.

Figure 25 shows the canonical correlation analysis spectrum. The top canonical correlation is 0.875, declining to 0.375 at the tenth component. The first five dimensions capture shared information between modality groups; the remaining dimensions capture private information specific to each modality group. The L2 norm of strategic private components correlates $r = 0.066$ with ϕ_S, providing a complementary signal: properties whose strategic embeddings carry more private (strategic-specific) information score higher on the prompted misalignment.

P Interaction Types

Figure 26 assigns properties to interaction types following Yu et al. [24]. The distribution: Synergy 30%, Redundancy 21%, Unique-S 14%, Unique-O 12%, Neither 23%. Misalignment matters most for synergistic and unique-S interactions, least for redundant ones, consistent with the framework: when modalities carry complementary or unique information, the strategic distortion has more room to operate. Redundant interactions, by definition, mean the strategic and objective channels carry the same content, which leaves little room for the strategic channel to distort.

Q The No-Photo Listings as a Negative Control

Figure 27 reports the no-photo listings analysis. Of 23,498 properties, approximately 782 have no listing photographs. These sell better, not worse: the bidding-war rate is 44.1% for no-photo listings versus 38.9% for photographed listings. Text-only misalignment for no-photo listings has an opposite-signed correlation with the price gap. The pattern is consistent with selection: agents who skip photos may do so for properties that sell themselves (a hot listing in a desirable neighborhood needs less marketing scaffolding) rather than for properties that need help. Excluding the 782 no-photo listings from the main analysis does not change any of the headline numbers; their pattern is a useful negative control. When curation drops out, so does the misalignment-outcome relationship.

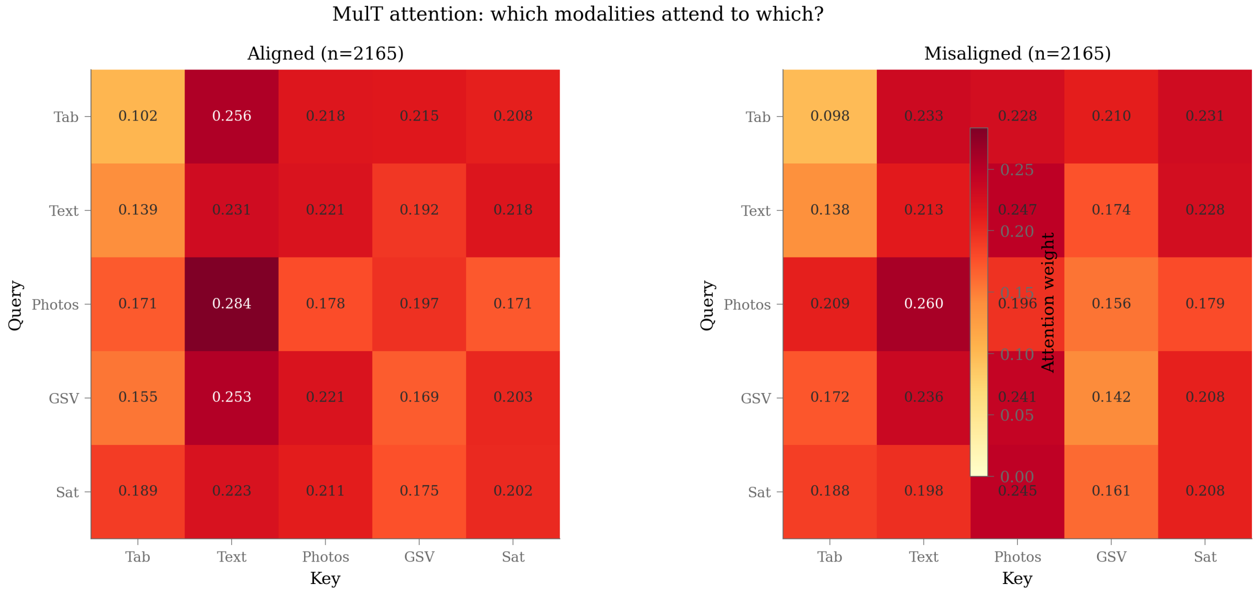

Figure 22: Cross-modal attention by alignment class. For misaligned listings, tabular queries attend more to photos and less to text.

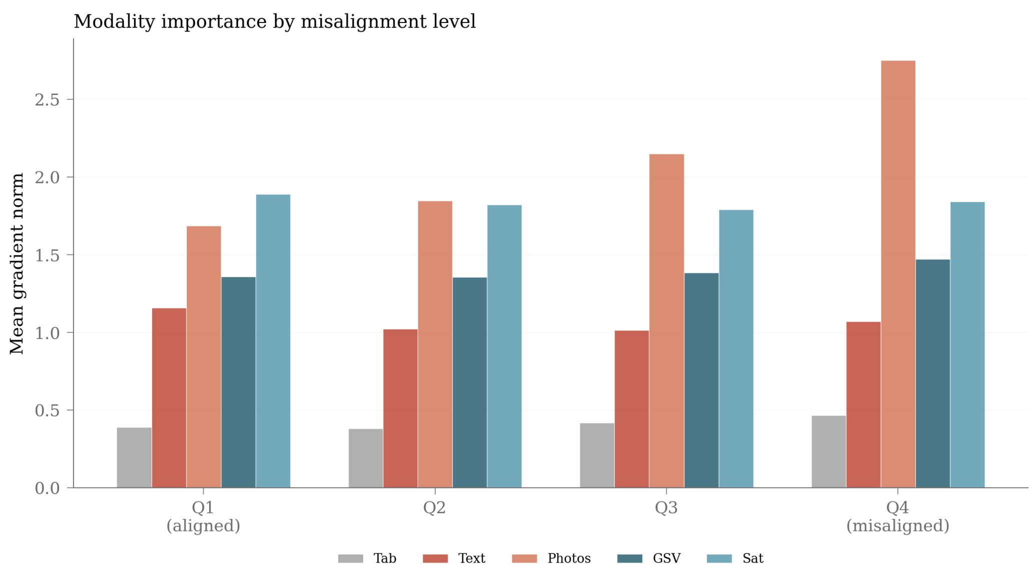

Figure 23: Gradient-based modality importance by misalignment quartile. Photo gradient norm rises monotonically from Q1 to Q4; tabular gradients are flat.

R Failed Alternatives and the Kamenica-Gentzkow Negative Result

Figure 28 compares four approaches. A Ridge model trained to predict residualized log price per square foot produces a misalignment score that correlates $r = -0.020$ with the price gap, in the wrong direction. A calibrated MLP yields $r = -0.034$; a contrastive InfoNCE model yields $r = -0.053$. The prompted hand-crafted method is the only one of the four producing a positive cross-sectional correlation ($r = 0.089$). These three value-supervised methods all target the value axis: what properties are worth. They miss the effort axis: how aggressively the agent presents the property. The supervised-on-gap method (evaluated separately as Δ^{learn} throughout the main paper) is a different construct again and recovers the data-driven adversarial axis.

Figure 29 shows the raw CLIP cosine distributions that motivated the projection onto a quality axis. The cross-class cosine has Pearson $r = 0.029$ ($p = 0.41$) with the price gap. Cosine measures topical content rather than quality agreement: a kitchen description has low cosine with a Street View facade regardless of accuracy. Strategic-objective pairs show wider variance than objective-objective pairs, reflecting variable agent curation, but the variance does not map onto market outcomes. This is the empirical motivation for projecting onto a quality axis instead of comparing embeddings directly.

Cross-modal disagreement. A learnable cross-modal disagreement metric of the form $\|\text{proj}_{\mathcal{S}}(X_{\mathcal{S}}) - \text{proj}_{\mathcal{O}}(X_{\mathcal{O}})\|^2$, fit end-to-end to predict Y^{gap}, is null: $r = 0.012$ with the supervised direction, $r = -0.023$ with the price gap, BW Q1/Q4 ratio $1.02\times$, within-building FE $p = 0.14$. The signal lives in specific structured directions of disagreement, not in overall magnitude.

Cross-city Kamenica-Gentzkow. The KG framework predicts that more rational buyers should attract less signaling. Per-city estimates of buyer rationality $\hat{\gamma}_c$ and mean signaling intensity $|\bar{\phi}_{\mathcal{S},c}|$ across 15 cities yield $r(\hat{\gamma}, |\bar{\phi}|) = -0.26$, opposite the predicted sign. The pattern is

Figure 24: Visual modalities add more value when listings are misaligned. Incremental R^2 per modality over a tabular baseline, split by alignment class.

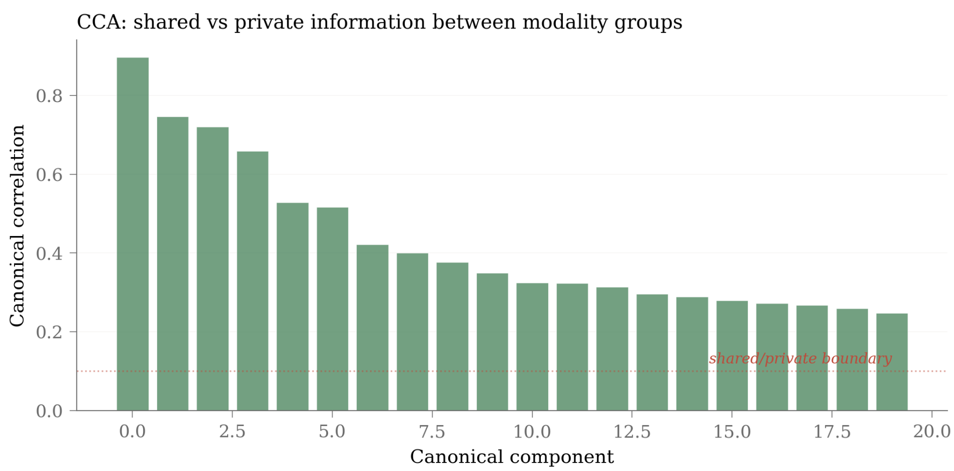

Figure 25: Canonical correlation between strategic and objective embedding blocks. Top correlation 0.875; first five dimensions capture shared information.

dominated by working-class lower-priced markets (Brockton, Lowell) having both higher property heterogeneity and higher signaling intensity. Signaling intensity tracks property heterogeneity, not buyer rationality.

S Spatial Distribution of Misalignment and Outcomes

Figure 30 maps misalignment and price gap geographically. Misalignment concentrates in downtown Boston luxury neighborhoods (Back Bay, Beacon Hill, Seaport) and affluent suburbs (Newton, Brookline). The luxury concentration is structural: high-end listings have more degrees of freedom in presentation and operate in markets where aspirational language is normative. Price gap shows a correlated but noisier spatial pattern. The two maps are not identical: misalignment-rich neighborhoods do not always show negative price gaps, because in those neighborhoods aspirational presentation is the norm and buyers have learned to discount it. The discrepancy between the two maps is itself evidence that misalignment captures *ex ante* strategic intent rather than *ex post* pricing outcomes.

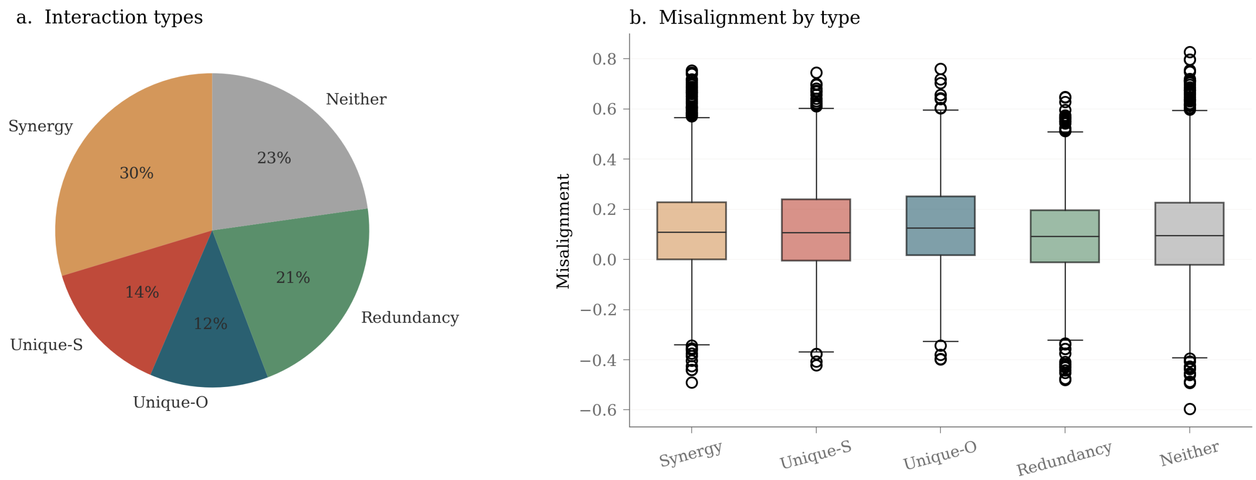

Figure 26: Interaction-type assignment. (a) Synergy 30%, Redundancy 21%, Unique-S 14%, Unique-O 12%, Neither 23%. (b) Misalignment by interaction type.

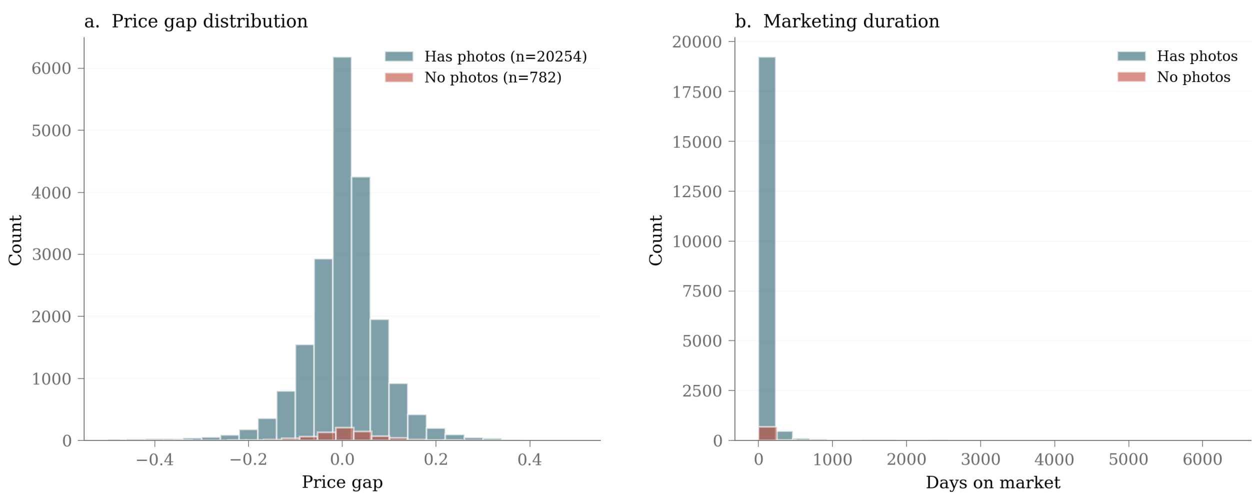

Figure 27: No-photo listings (red, $n = 782$) compared to photographed listings (teal, $n = 20{,}254$). Price gap distribution and marketing duration.

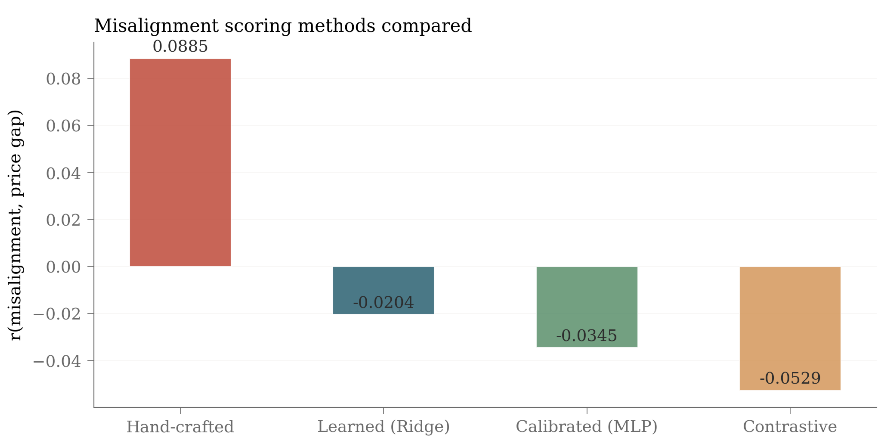

Figure 28: Value-supervised alternatives produce wrong-signed scores. Only the prompted method recovers the correct sign on price gap.

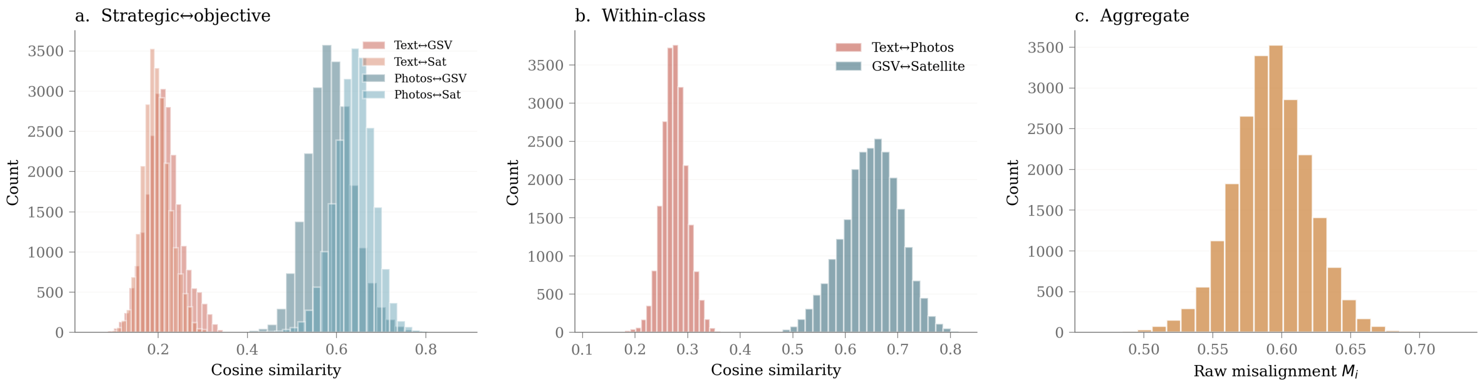

Figure 29: Raw CLIP cosine fails to predict outcomes. (a) Strategic-objective cosines; (b) within-class cosines; (c) aggregate raw misalignment.

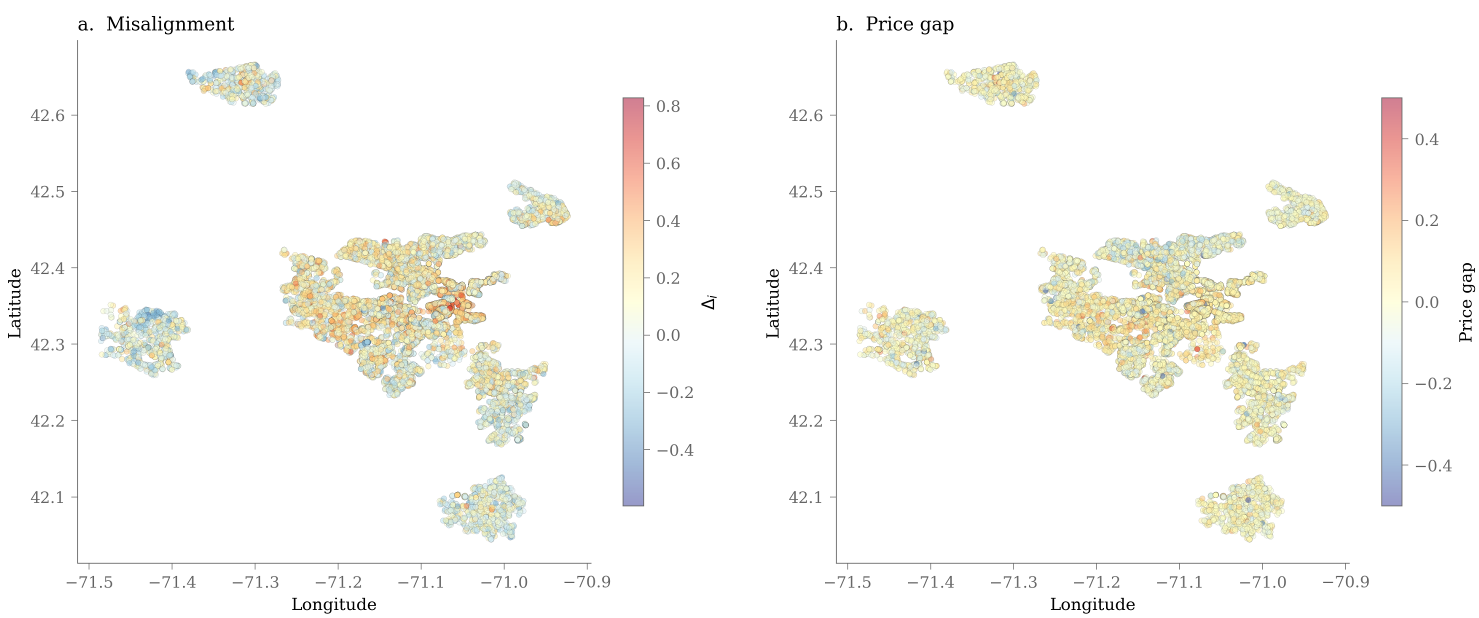

Figure 30: Spatial distribution of misalignment and price gap across the greater Boston area. (a) Misalignment Δ_i concentrates in downtown Boston luxury neighborhoods and affluent suburbs. (b) Price gap shows a correlated but noisier spatial pattern.